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Climate risk and firms' R&D investment: Evidence from China

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ABSTRACT

This paper investigates the impact of regional climate risk on firms' R&D investment using data from 23854 firm-year in China from 2008 to 2021. We find that regional climate risk has a significantly negative impact on firms' R&D investment, which shows that climate risk inhibits firms' innovation. In addition, the above-mentioned effect is more pronounced in firms with lower cash holdings and lower manager shareholdings. The mechanism analysis shows that climate risk reduces firms' R&D investment by lowering firms' profitability and exacerbating financing constraint. Further analysis based on China's institutional environment also shows that the above impact of climate risk on firms' R&D investment is more pronounced for firms with non-state property right and lower government subsidies.

1. Introduction

Climate change has become one of the most pressing issues today (Dutta et al., 2023). The global average temperature has risen by 1 °C compared with pre-industrial times. Global temperatures are still on an upward trend and are expected to rise by 2–4 °C by 2100 (Hong et al., 2019). Climate change has seriously affected natural ecosystems and economic development (Venturini, 2022). Climate change poses enormous climate risk. Climate risk refers to the uncertainty of economic and financial activities caused by extreme weather, natural disasters, global warming, other climatic factors, and the social transition to sustainable development.[2] Researchers have studied the economic impact of climate risk from the following perspectives. First, many researchers have examined the macro-level economic impact of climate risk, such as economic growth (Zhao et al., 2022), employment (Balvers et al., 2017), monetary policy (Dafermos et al., 2018) and fiscal revenues (Yang & Tang, 2022). Second, from the perspective of the medium-level, some researchers have examined the effect of climate risk on market efficiency (Hong et al., 2019), financial stability (Pagnottoni et al., 2022), and stock market volatility (Chen et al., 2023; Lv & Li, 2023). Third, from the perspective of the micro-level, the existing literature has examined the impact of climate risk on corporate capital structure (Ginglinge and Moreau, 2019), risk-taking (Xu et al., 2022), corporate social responsibility (Mbanyele & Muchenje, 2022) and green investment (Dutta et al., 2023).

Innovation is an important strategy for sustainable development. Firms' R&D investment is the fundamental engine for creating sustainable innovations (Sarpong et al., 2023). However, firms' R&D investment is subject to different risk factors (Evans & Guthrie, 2004). The existing research about the impact of risk on firms' R&D investment mainly concerns market risk (Luo et al., 2022), political risk (Yang et al., 2022), legal risk (Guan & Zhang, 2021), financial risk (Wen et al., 2023) and operational risk (Beladi et al., 2021). The existing literature about the impact of risk on firms' R&D investment is relatively abundant. However, less has explored whether and

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[2] See IPCC-Intergovernmental Panel on Climate Change for more details on climate risk.

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how climate risk affects firms' R&D investment. This paper fills in the above gap.

This paper selects Chinese firms as research samples to explore the impact of regional climate risk on firms' R&D investment and its influence mechanism, which is of great significance to China's economic development. China is implementing an innovation-driven development strategy to promote social and economic development and green transformation by enhancing innovation capacity and efficiency. On February 29, 2024, *the National Bureau of Statistics of the People's Republic of China* released the Statistical Bulletin of the People's Republic of China on National Economic and Social Development in 2023, which shows that China's R&D investment will reach 3327.8 billion yuan in 2023, with an intensity (ratio to GDP) of 2.64 %. As shown in Fig. 1 below, although the intensity of China's R&D investment shows a continuously increasing trend in the past years but it is far below the 5 % level of innovative countries[3]. Firms are the core subjects of social R&D innovation. Therefore, it is very important for us to identify various obstacles and challenges faced by Chinese firms in the process of R&D innovation so as to take targeted business measures and policy interventions. That is of great significance for the effective implementation of China's innovation-driven development strategy (Kong et al., 2024).

Meanwhile, there are also some advantages to conducting such research based on Chinese firms. First, as the country with the second largest population in the world, China has extremely complex geographical and climatic conditions. The climate risk faced by China is widely representative. China not only has a vast territory and spans multiple climatic zones, but also is significantly affected by monsoons, frequent extreme weather and climate events (Ren et al., 2023). Fig. 2 below shows the natural disaster-affected area data for each province of China in 2022, demonstrating the differences in climate risk across regions. This complex climate situation makes China an important and representative case country for global climate change research. Secondly, studying the impacts of China's climate risk and relevant policy interventions will not only help to improve the domestic public's understanding and awareness of China's climate risk, but also provide useful references and lessons for other countries and promote the progress and development of global climate governance. Finally, this paper also has the empirical advantage in examining the impact of regional climate risk on the firms' R&D investment. Current research mainly used a measurement indicator of country-level climate risk to study the impact of climate risk on firms' business behavior (Huang & Lin, 2022; Xu et al., 2022; Song et al., 2023). However, each country has different political economy, legal system, culture and social norms. These country-specific characteristics will inevitably affect the robustness of empirical test results when using cross-country samples. In addition, the above empirical analysis faces the challenges of different data standards and statistical caliber in different countries. In contrast, studying the impact of regional climate risk of China on firms' R&D innovation not only ensure the consistency and comparability of empirical data, but also helps to eliminate the interference of country-specific characteristics on the relationship between climate risk and firms' R&D investment.

Thus, referring to Hong et al. (2019), Pankratz et al. (2023), Pankratz and Schiller (2024), and Marotta and Ciccarelli (2024), this paper uses annual mean temperature (*Amt*), annual mean precipitation (*Amp*), annual mean relative humidity (*Amrh*), annual sunshine duration (*Asd*), and total CO_2 emissions per unit of GDP (*IE*) of China's prefecture-level cities, and design the indicator to measure regional climate risk. Then, we use Chinese A-share listed firms from 2008 to 2021 as research samples to explore the impact of regional climate risk on firms' R&D investment. The results show that climate risk reduces firms' R&D investment. We further examined the moderating effect of cash holdings and manager shareholdings on the relationship between climate risk and firm R&D investment. It is found that high cash holdings and manager shareholdings can alleviate the negative impact of climate risk on firm R&D investment. The analysis of the influence mechanism verifies that climate risk has a negative impact on firms' R&D investment by reducing firms' profitability and exacerbating firms' financing constraint. In addition, based on the characteristics of China's institutional environment, this paper also analyzes the differences of the above impacts in different types of property rights (state-owned vs. non-state-owned) and the level of government subsidies. We find that the impact of climate risk on firm R&D investment is more significant in firms with non-state property right and lower government subsidies.

To the best of our knowledge, the marginal contribution of this paper mainly shows in the following three aspects. Firstly, this paper examines the impact of climate risk on firms' R&D investment and provides empirical evidence based on data from Chinese listed firms. Existing research has examined the impact of the climate risk on corporate capital structure (Ginglinge and Moreau, 2019), risk-taking (Xu et al., 2022), CSR (Mbanyele & Muchenje, 2022), and green investment (Dutta et al., 2023). This paper verifies the impact of climate risk on firms' R&D investment in China and identifies its mechanism. The finding supplements the research about the micro-level economic impact of climate risk and helps to understand the influence mechanism of climate risk on firms' R&D investment.

Secondly, this paper enriches the research about the impact of risk factors on firms' R&D investment. Existing research mainly involved market risk (Luo et al., 2022), political risk (Yang et al., 2022), legal risk (Guan & Zhang, 2021), financial risk (Wen et al., 2023) and operational risk (Beladi et al., 2021). Few studies focus on the impact of climate risk on firms' R&D investment as we know. This paper studies the impact of the climate risk on firms' R&D investment, which is a supplement to the research on the risk factors influencing firms' R&D investment.

Thirdly, this paper identifies the moderating effects of internal corporate characteristics (i.e., cash holdings level and manager shareholdings level) and external institutional environmental factors (i.e., state owned property attribute and government subsidies level) on the relationship between climate risk and firms' R&D investment. This not only provides empirical evidence and directions

[3] See more details for https://www.gov.cn/lianbo/bumen/202402/content_6934935.htm.

[4] The data used in this Fig. 1 from *the National Statistical Bulletin on Science and Technology Investment* and *Statistical Bulletin of China's National Economic and Social Development*.

[5] Fig. 2 shows the disasters of China's provinces in 2022 (not including Taiwan, China). The higher the disaster-affected area of the province, the darker the color of the province in Fig. 2 is. The data are from the Ministry of Emergency Manager of China.

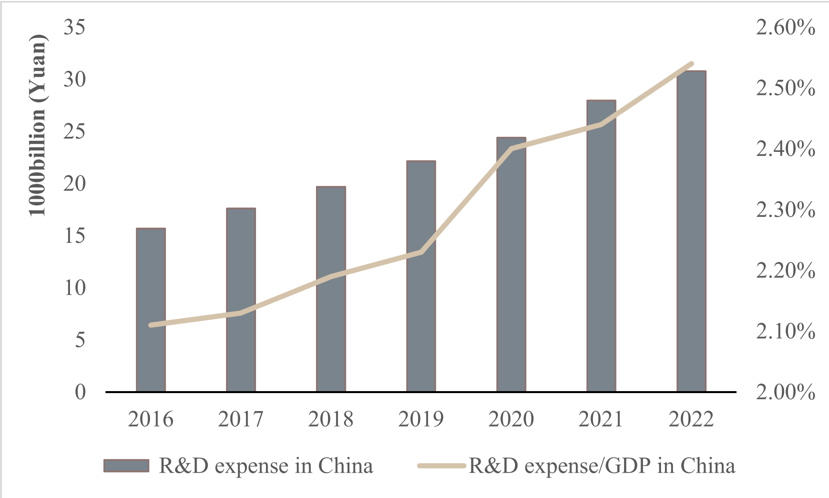

Fig. 1. Trend of China's R&D investment intensity from 2016 to 2022[4].

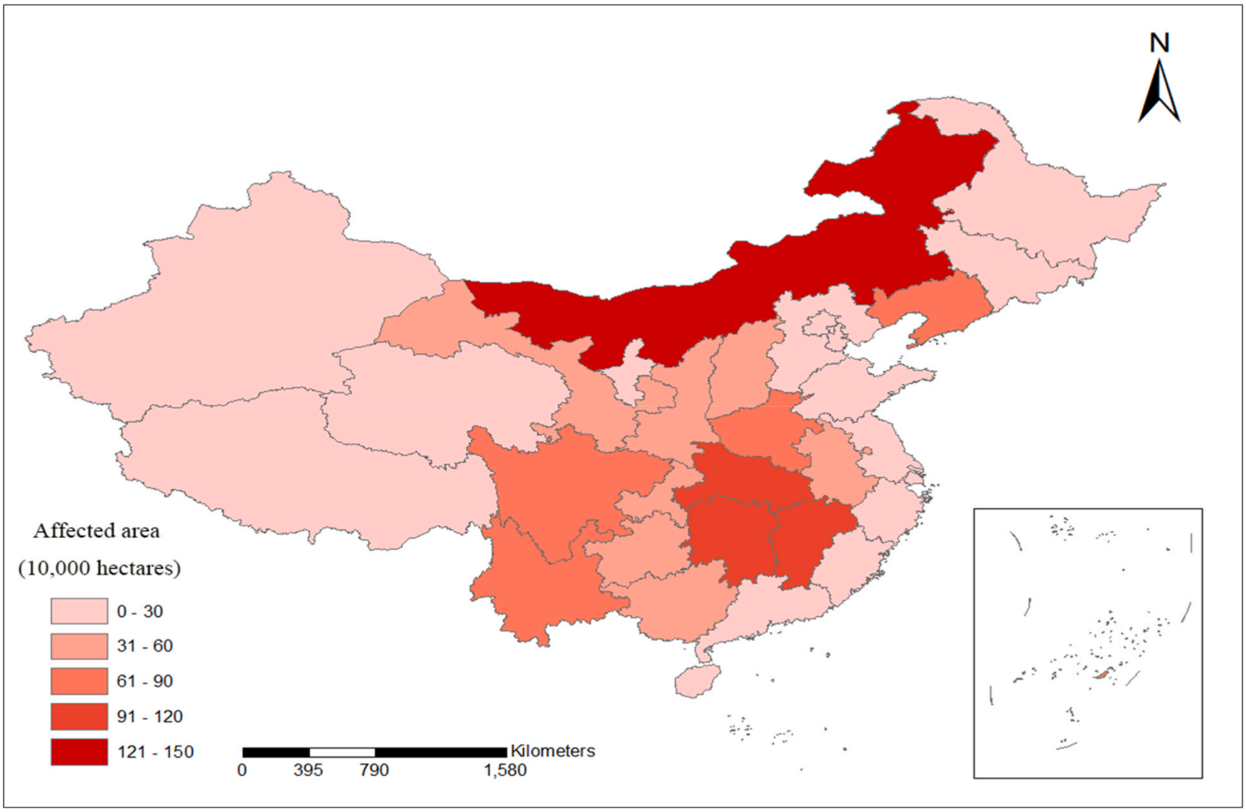

Fig. 2. The disasters-affected area in 2022 b y province in China.[5].

for firms' coping strategies to address climate risk, but also has important policy implications. Findings on the moderating effect of state ownership and government subsidies have implications for other countries also with prevalent state ownership and government subsidies, such as Finland, Austria, Singapore, and Malaysia (Wang et al., 2017).

The remainder of the paper is organized as follows. Section 2 gives the literature review and hypothesis development. Section 3 describes the sample collection and the empirical method. Section 4 presents the results. Section 5 provides the conclusions.

2. Literature review and hypothesis development

2.1. Literature review

2.1.1. The impact of climate risk on firms

Climate risk was first proposed by James Hansen in the 1980s who was concerned about the impact of carbon dioxide emissions on the greenhouse effect (Venturini, 2022). Environmentalists and meteorologists then began studying the impact of climate change on the ecological environment. Stern and Taylor (2007) further extended the impact of climate risk to economics for the first time and analyzed the impact of climate risk on social wealth. In addition, Stern and Taylor (2007) have studied the impact of climate risk on the financial industry, which deepens the study of climate risk's influence in the field of economics. As the macro-economy and medium-industry's micro subject, firms are also affected by climate risk. Therefore, the impact of climate risk on firms has also attracted the attention of researchers.

Existing research about the impact of climate risk on firms mainly focuses on the following two aspects. Firstly, some literatures studied the impact of climate risk on firms' performance. Addoum et al. (2020) and Ding et al. (2021) found that climate risk reduces firms' revenue and performance. Sun et al. (2020) analyzed the impact of climate risk on China's mineral-listed companies and found that climate risk would have a negative effect on the financial performance of mining companies. Dutta et al. (2023) concluded that climate risk increases firms' operating costs and has a negative impact on firms' revenue. Secondly, some researchers studied how firms responded to climate risk, such as earnings management behavior (Ding et al., 2021), risk-taking behavior (Xu et al., 2022), and risk management behavior (Mbanyele & Muchenje, 2022).

In order to measure the climate risk faced firms, existing studies have mainly constructed the measurement indicators of climate risk from the country, regional and firm levels. At the country level, Hong et al. (2019) used the Palmer Drought Severity Index (PDSI), which reflects information that temperature and soil moisture of the country, to measure climate risk. Among the natural disasters that can be amplified by climate change, drought is considered one of the most damaging to economic activity. Zhao and Parhizgari (2024) used the Climate Risk Index (*CRI*) as measure indicator for climate risk. *CRI* is published by Germanwatch, and is an annual score for each country, which reflects information on the annual economic losses of countries affected by extreme weather. At the regional level, Huynh and Xia (2021) used the economic loss of natural disasters in the regions where the firms and their branches are located to measure climate risk. Pankratz et al. (2023) used the temperature change in the region where the firm is located to measure climate risk. Pankratz and Schiller (2024) further used temperature and water-level changes to measure climate risk. Xie and Li (2024) measured climate risk at the city level using the level of temperature anomalies. This indicator represents anomalies in urban temperatures relative to historical levels. At the firm level, Sautner et al. (2023) developed a method to measure firms' climate risk, calculating the proportional change in the text content related to climate risk and responses in the firms' conference calls and annual financial reports. Fatima and Ziane (2024) measured climate risk using the number of climate-related risks mentioned by firms in the CDP questionnaire multiplied by their likelihood of occurrence, magnitude of impact, and time frame.

The climate risk indicators of the country and regional levels mainly use indicators directly related to climate change, which can directly reflect the real situation of climate change and help to assess the actual situation and impact of climate risk more accurately. In addition, the country and regional level indicators are based on actual climate-related data, which makes the assessment results more objective and may reduce the interference of subjective judgments on the assessment results.

2.1.2. The influencing factors of firms' R&D investment

The existing literature has conducted in-depth research on the influencing factors of firms' R&D investment and achieved abundant results. Much literature has studied the impact of external law and regulation (Li et al., 2022; Tran, 2023), credit policy (Zhang & Kong, 2022), firm cross-section characteristics (Beladi et al., 2021), internal individual characteristics of manager (García et al., 2023) on firms' R&D investment.

Risks and uncertainties will affect the investment decisions of firms (Pindyck, 1991). R&D is a high-risk investment activity. Therefore, risk factors have an important impact on firms' R&D investment. From the perspective of firms' external risk factors, the existing research has studied the impact of market risk (Luo et al., 2022), political risk (Yang et al., 2022), and legal risk (Guan & Zhang, 2021) on firms' R&D investment. Luo et al. (2022) found that firms with higher political exposure and political risk spend less on R&D investment. Yang (2023) documented that firms with strong employment protection legislation have lower R&D expenditure and investment efficiency. Firms' internal risk factors also have a significant impact on R&D investment, including operational risk (Beladi et al., 2021) and financial risk (Wen et al., 2023). Beladi et al. (2020) found that firms with high operational risk reduce R&D investment as if they are more risk-averse in investment decisions. Wen et al. (2023) examined the effect of financial risk on green innovation and found that financial risk has a significant negative impact on green innovation by decreasing patent applications related to environmental management technologies.

2.2. Theoretical analysis and hypothesis development

2.2.1. Climate risk and firms' R&D investment

Firms' R&D investment is characterized by high risks, continuous inputs and irreversibility (Beladi et al., 2021). Firms' R&D investment often requires adequate financial support. Climate risk has an adverse impact on business operations, increase their business operating costs and reduce their profitability (Ginglinger & Moreau, 2023), which will weaken firms' internal financial support for its R&D investment. Firstly, climate risk affects the production process of firms and increase firms' operating costs. Floods, hurricanes, and other extreme weather events can directly damage business equipment and infrastructure, resulting in increased equipment repair and replacement costs (Ginglinger & Moreau, 2023). In addition, climate change exacerbates the volatility of energy and raw material prices, increasing procurement costs for firms (Dutta et al., 2023). Secondly, adverse weather environment may disrupt key links in the supply chain, leading to supply chain disruptions (Ginglinger & Moreau, 2023). Natural disasters can damage roads, bridges, railways, and airports, causing delays or disruptions in the transportation of raw materials and products. This will result in an increase of transportation costs, including transportation delays, route changes and other additional costs. (Cheng et al., 2023). Finally, climate risk increases the cost of insurance for firms. Faced with higher climate risk, firms need pay higher insurance premiums to cover potential losses (Ginglinger & Moreau, 2023). In brief, by increasing firms' operating costs, climate risk force firms to invest more funds, resources and manpower into production and operation activities to offset the negative impact of climate risk and ensure the continuous operation of core business (Yan et al., 2024). Thus, funds originally used for R&D may be cut or reallocated, crowding out the firms' R&D investment. At the same, rising costs caused by climate risk reduces firms' profitability and weakens the internal financing support ability for firms' R&D investment.

Climate risk will also exacerbate the external financing constraint of firms and then decrease firms' R&D investment. According to the risk pricing theory, the climate risk increases the systemic risk of firms. Investors will demand a higher risk premium to compensate for the higher climate risk, and leads to an increase in external financing costs of firms (Charpentier, 2008). In addition, climate risk may trigger business difficulties and increases the debt default risk of firms. Banks will adjust their credit policies and adopt stricter credit conditions for firms with higher climate risk, such as raising loan interest rates, lowering loan quotas or requiring firms to provide more guarantee measures, which will worsen the external financing constraint of firms (Deng et al., 2024). Thus, it is difficult for firms to provide adequate financial support for their R&D investments through external financing when facing climate risk.

Finally, climate risk may also have a negative impact on manager's motivation for R&D investment. Extreme weather events may trigger risk-averse behavior by managers. According to agency theory, manager is often considered to be risk-averse (Addoum et al., 2020). Due to climate risk will have a negative impact on firms' business activities and increase their business risks (Sun et al., 2020), manager in the firms with high climate risk is more inclined to adopt conservative innovation strategies and reduce R&D investment (Sendstad and Chronopoulos, 2020).

To sum up, climate risk will weaken the financial ability for firms' R&D investment by reducing the profitability of firms and intensifying firms' financing constraint and inhibit the motivation of managers to invest in R&D, thus leading to a decrease of the firms' R&D investment. Therefore, we propose the following H1.

H1. Climate risk has a negative impact on firms' R&D investment.

2.2.2. *Moderating effect of firm cash holdings*

The precautionary motive theory of corporate cash holdings suggests that firms maintain a certain level of cash reserves to respond to emergencies and external risks such as market volatility, economic downturns, and unexpected events, ensuring their continued operations and sustainable growth (Han & Qiu, 2007). In the event that extreme climate events disrupt a firm's supply chain and production activities, constraining R&D investment, sufficient cash reserves can provide the firm with the financial flexibility to manage the pressures of supply chain adjustments. Moreover, these cash reserves allow for swift implementation of emergency measures, such as repairing damaged facilities or adjusting production plans, thereby preventing production interruptions and market supply shortages. This, in turn, mitigates the negative impact of climate risk on the firm's R&D and innovation efforts.

In addition, Brown and Petersen (2011) found that corporate cash reserves help smooth R&D investments, ensuring the continuity and stability of a firm's R&D activities, which is reflected in the simultaneous growth of cash holdings and R&D investment for young companies engaged in R&D. Sufficient cash reserves not only provide direct financial support for a firm's innovation effort but also contribute to maintaining financial stability, thus mitigate the decline of firms' R&D investment capacity due to financing constraint caused by climate risk.

Therefore, it is reasonable to expect that the impact of climate risk on firms' R&D investment is less significant in firms with higher cash holdings. Based on the above analysis, we propose the following H2.

H2. With the increase of firms' cash holdings, the negative impact of climate risk on firms' R&D investment becomes weaker.

2.2.3. *Moderating effect of manager shareholding*

Based on the agency theory, due to managers cannot fully diversify the risks faced by their personal wealth portfolio, they tend to avoid risks and have short-term behaviors when making investment decisions (Wright et al., 1996). Furthermore, due to the extreme risk aversion preference, managers are reluctant to invest in R&D activities with high-risk nature (Addoum et al., 2020). In order to restrain managers' opportunistic behavior and change managers' risk aversion, shareholders often design incentive mechanisms to reduce managers' conservatism in investment decisions (Chen & Ma, 2011; Fu et al., 2024). The existing research shows that the equity incentive method of managers' shareholding is a typical long-term incentive method adopted by firms, which helps to realize the benefit sharing and risk sharing between shareholders and managers, and helps managers to overcome the short-term behavior tendency (Wu et al., 2022). In addition, equity incentive can also change managers' risk preferences and improve managers' risk acceptance (Wruck & Wu, 2021). In view of this, when firms are faced with climate risk, equity incentives help managers to pay more attention to the sustainable development and long-term competitiveness of firms, and more actively invest in R&D and innovation of low-carbon technologies to cope with the pressures brought by climate change.

Therefore, if climate risk leads to a decline in firms' R&D investment partly by weakening managers' motivation for R&D investment, it can be expected that the impact of climate risk on firms' R&D investment is more significant in firms with lower manager shareholdings. Therefore, we propose the following H3.

H3. With the increase of manager shareholdings, the negative impact of climate risk on firms' R&D investment becomes weaker.

3. Data and summary statistics

3.1. Sample

This paper employs the data of 3335 Chinese A-share listed firms from 251 prefecture-level cities in China. On February 15, 2006, the Ministry of Finance of the People's Republic of China issued the new *China Accounting Standards for Business Enterprises* (CAS) by order No.33, and required listed companies to formally implement them on January 1, 2007. The CAS requires listed companies to disclose the amount of R&D investment in detail in the notes of their financial statements. In order to maintain the consistency of statistical caliber of data, this paper selects samples from 2008 to 2021 for research. After excluding the observation samples that are in the financial industry and have the marks of ST and ST*, we obtain a sample of 23854 firm-year observations. Appendix A presents a list of the 251 prefecture-level cities and the number of firms in each city. Appendix B presents the distribution of our sample on the basis of year and industry.[6] The firm-level data come from the CSMAR database. The data related to prefecture-level cities come from *China Statistical Yearbook*, *China City Statistical Yearbook*, and *China Environmental Statistical Yearbook*.

3.2. Variables

3.2.1. Dependent variable (RD)

The dependent variable is the firms' R&D investment (*RD*).

This paper employs two proxy variables to measure firms' R&D investment. Based on existing literature (Wen & Zhao, 2020; Chi et al., 2023), the first proxy variable is *RD_or*, which is measured as the ratio of firms' R&D investment to its operating revenues, and the variable is multiplied by 100 in the empirical analysis to improve the readability of the regression coefficient (Tribo et al., 2007; Wen & Zhao, 2020). The second proxy variable is $RD\_asset_{i,t}$, which is measured as the ratio of firms' R&D investment to its total assets (Chi et al., 2023). The variable ($RD\_asset_{i,t}$) is also multiplied by 100.

3.2.2. Independent variables (cr)

The independent variable is regional climate risk (*Cr*).

Previous studies on the measurement of climate risk mainly divided into two methods: single indicator measurement (Marotta & Ciccarelli, 2024; Xie & Li, 2024) and multi-indicators comprehensive evaluation model (Ginglinger & Moreau, 2023). Climate risk is a complex and multidimensional concept, involving physical risk (such as extreme weather events, sea-level rise, temperature change, etc.), and economic transition risk, among others.[7] The single indicator measurement captures only partial characteristics of climate risk, failing to reflect it comprehensively and systematically. In contrast to single indicator measurement model, the multi-indicators comprehensive evaluation method fully considers the multitude of factors that influence economic activities caused by climate change, and is more effective in evaluating climate risk.

Climate risk can be divided into the physical and transitional risk (Carney, 2015). The physical risk refers to the mainly negative impact of climate change and weather-related events on company operations, society and supply chains (Venturini, 2022). The transitional risk refers to all the possible scenarios coherent with a path to a low-carbon economy and all related implications for fossil fuels and dependent sectors (Curtin et al., 2019). The physical risk is directly related to climate change, including temperature, precipitation changes and other phenomena. Among the measurement indicators of physical risk, temperature change is the main indicator used, followed by drought (Venturini, 2022). Changes in temperature, sunshine duration, precipitation, and humidity all contain information about drought.

Therefore, based on relevant literature (Hong et al., 2019; Marotta & Ciccarelli, 2024; Pankratz et al., 2023; Pankratz & Schiller, 2024; Xie & Li, 2024), this paper selects five secondary indicators to construct measurement indicator of regional climate risk, including annual mean temperature (*Amt*), annual mean precipitation (*Amp*), annual mean relative humidity (*Amrh*), annual sunshine duration (*Asd*) and total CO_2 emissions per unit of GDP (*IE*).[8] Among them, the first four indicators are used to reflect the physical risk, and the last indicator is used to reflect the transition risk.

Temperature change is one of the important manifestations of climate system change, which has a profound impact on natural ecosystems and economic activities (Campbell & Diebold, 2005). Temperature change will lead to the increase of energy consumption and production cost of firms (Sendstad & Chronopoulos, 2020). Extreme weather events related to temperature changes (such as

[6] According to the *Value Creation Report of China's Manufacturing Listed Companies in 2021*, as of December 31, 2021, the number of listed companies in China's manufacturing industry accounted for 66.7 % of the number of listed companies in China. Due to the high proportion of manufacturing firms in the research sample, this paper tests whether there is a difference in the negative impact of climate risk on firms' R&D investment between manufacturing and non-manufacturing firms in groups. The results are presented in Appendix E. The results show that there is no significant difference in the negative impact of climate risk on firms' R&D investment between manufacturing and non-manufacturing firms.

[7] See more details for https://www.bankofengland.co.uk/legal/privacy-and-cookie-policy.

[8] It is undeniable that climate risk is also related to other factors, such as sea level rise, the application of clean energy technologies, and low-carbon policies. Limited to the availability of relevant data, we have not incorporated these factors into the design of climate risk indicator.

drought) can also lead to production interruption, equipment damage and labor costs, affecting the normal operation of firms (Ginglinger & Moreau, 2023).

Precipitation is an important part of the ecosystem, and the variation of precipitation reflects the dry and wet conditions of the climate and the frequency of extreme weather events (Tabari, 2020). Extreme changes in precipitation (such as heavy rains, floods and droughts) can affect infrastructure and production equipment, which in turn affects business operations.

In addition, extreme changes in temperature and precipitation can also affect logistics and transportation, leading to supply chain disruptions and raising operating costs for firms (Ginglinger & Moreau, 2023).

The change of relative humidity will affect the production and operation of firms. First, for the production environment that needs precise control of humidity (such as precision instrument manufacturing, electronic industry, etc.), the change of relative humidity will directly affect the accuracy of equipment, reduce the life of production equipment, and then lead to the decline of product production quality. Second, humidity changes will affect the quality and shelf life of raw materials. The quality of products produced and stored in an environment with unsuitable humidity is also affected.

Changes in sunshine duration also have a significant impact on firms. Sunshine hours directly affect energy prices and the energy consumption of firms, which in turn affects their production costs.[9] In addition, for agricultural firms, changes in sunshine duration directly affect their product quality and yield (Zhang et al., 2017).

Under China's "carbon peaking and carbon neutrality goals", the real economy will be profoundly affected by the implementation of climate policies, shifting consumer preferences, and the application of new technologies in the transition to a low-carbon economy (Li & Pan, 2022). The extent to which firms are affected by climate transition risk depends primarily on CO_2 emissions.[10] Therefore, CO_2 emissions per unit of GDP can be used to measure climate transition risk (Marotta & Ciccarelli, 2024).

The secondary indicators mentioned above are standardized and absolute values are taken to make them dimensionless and comparable. Standardized secondary indicators not only meet the requirements of comparability, but also can describe the fluctuation degree of climate change (Hong et al., 2019; Xie & Li, 2024).

The specific calculation method is as follows:

$$M^{W}_{Var_{i,t}} = \sum_{k=1}^{W} Var_{i,t-k} \Big/ \mathrm{W} \tag{1}$$

$$SD^{W}_{Var_{i,t}} = \sqrt{\sum_{k=1}^{W} \left(Var_{i,t-k} - M^{W}_{Var_{i,t}}\right)2 \Big/ \mathrm{W}} \tag{2}$$

$$Var^{W}_{i,t} = \left|Var_{i,t} - M^{W}_{Var_{i,t}}\right| \Big/ SD^{W}_{Var_{i,t}} \tag{3}$$

Where Var is the annual mean temperature (*Amt*), annual mean precipitation (*Amp*), annual mean relative humidity (*Amrh*), annual sunshine duration (*Asd*) and total CO2 emissions per unit of GDP (*IE*) of the prefecture-level city. i in year t. W is the number of lag window years and is assigned to 5. Finally, the principal component analysis was applied in measuring regional climate risk. The KMO value is 0.790, and P values of the Bartlett test of sphericity are 0.000. The combined score is used to measure regional climate risk (*Cr*). The higher *Cr* is, the greater the regional climate risk is. The principal component structure of regional climate risk is shown in Appendix D.

In order to further verify the rationality and effectiveness of the above indicator for measuring regional climate risk, this paper examines the correlation between the disaster-affected area indicator (*Ae*) and the climate risk indicator (*Cr*) in each province and year in China from 2008 to2021[11]. Among them, the climate risk indicator (*Cr*) of each province in each year is determined by calculating the annual average of the climate risk indicator of the prefecture-level cities contained in each province, and the disaster-affected area indicator (*Ae*) is the ratio of the disaster-affected area of each province to the total area of the province in each year[12]. We believe that a larger area affected by natural disasters indicates that a wider geographical area has been affected by natural disasters, which means that the climate conditions in the region may be more unstable or extreme, and the probability of being affected by climate extremes again in the future is higher. Meanwhile, the larger the impact range of natural disasters is, the more serious the economic loss, social impact and destruction of the natural environment will be. This further highlights the serious threat of climate risk to the region. Fig. 3 shows the scatter plot between the above two. It can be seen from the figure that there is a high degree of consistency and positive

[9] For firms that rely on solar power, an increase in sunshine duration means more solar resources are available, which may reduce power generation costs and improve energy efficiency. In addition, the change of sunshine duration will lead to an increase in the demand for energy such as heating in winter and cooling in summer, which will have an impact on the energy consumption of firms and the market energy price, and indirectly affect the operating costs of firms (Mulder & Scholtens, 2013).

[10] The Chinese government uses CO_2 emissions as the main criterion to evaluate the environmental performance of firms, and reward and punish the firms.

[11] Due to the availability of disaster data in each region and the visibility of data display, province-level data were used for verification.

[12] The annual disaster-affected area data of each province in China are from the Ministry of Emergency Manager of China.

[13] The disaster-affected area indicator (*Ae*) is the ratio of the affected area to the total area of each province in each year. The climate risk indicator (*Cr*) of each province is the annual average of the climate risk indicator of the prefecture-level cities under the jurisdiction of the province.

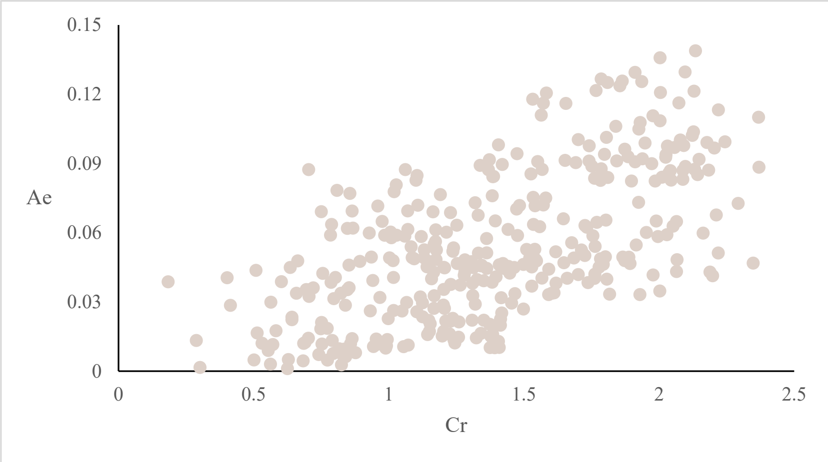

Fig. 3. Scatter plot of the climate risk (*Cr*) and the disaster area (*Ae*) of provinces in China from 2008 to 2021[13].

correlation between the climate risk indicator (*Cr*) of each province and the disaster-affected area of each province (the correlation coefficient is 0.725), which further proves the validity and rationality of the above climate risk indicators.

3.2.3. Moderating variable

The first moderating variable is cash holdings (represented by *Cash*), which reflects the level of cash held by the firms. According to Acharya et al. (2012) and Ye et al. (2024), we measured cash holdings (*Cash*) by the ratio of cash and marketable securities to total assets. The second moderating variable is manager shareholdings (represented by *Managshare*). Following previous studies (Fu et al., 2024; Wu et al., 2022), the ratio of directors, officers, and supervisors' total shareholdings number to firms' total share number is used to measure manager shareholding in this paper.

3.2.4. Control variables

According to the research of Khan et al. (2020) and Chi et al. (2023), we have controlled the variables that may pertain to R&D investments to address the possible heterogeneity. Firm size (*Size*) and firm age (*Age*) represent size and other factors that may affect the acquisition of firms' R&D resources. Debt to asset ratio (*Debt*) reflect the sustainability of internal sources of R&D funding and the level of firm financial resources. Ratio of fixed assets to total assets (*Fixed*) and Tobin Q (*Tbq*) reflect the asset structure and market valuation of the firm respectively. Growth rate of total assets (*AR*) reflects the investment growth profile of firms. Manager expense ratio (Fee), whether the Chairman of the Board and CEO are the same person (*Ceo*), and Number of the board of Supervisors (*BS*) reflect the manager efficiency and governance issues of the firm. Herfindahl index of income (*Hhi*) captures the degree of competition in the market. Total urban product divided by the number of people in the city (*Pgdp*) and urban industry structure (*UP*) reflect the economic level and industrial structure of the city that may affect firms' access to R&D resources. The specific control variables definition is shown in Appendix C.

3.3. Empirical model

In order to verify H1 to H3 in this paper, this paper conducts the Hausman test, and the results show that the null hypothesis is rejected. In addition, considering that the dependent variable *RD* has censoring phenomenon at 0, this paper uses both the Fixed effect model and Tobit model[14] to test the relationship between regional climate risk and firms' R&D investment, referring to the existing literature (Liu et al., 2023; Mertzanis et al., 2023; Nguyen et al., 2024). We construct the following models (4) to (6). Model (4) is used to examine H1. If β_1 is significantly negative, then H1 is supported. Models (5) and (6) are used separately to test H2 and H3. According to H2 and H3, the regression coefficients β_1 is expected to be significantly negative and the regression coefficients β_2 is expected to be

[14] The Fixed effect model can effectively solve the problem of missing variables, reveal the fixed effects and improve the estimation accuracy. However, due to the truncation phenomenon of dependent variable *RD* at 0, the Fixed effect model may have estimation errors. The Tobit model can estimate the regression coefficients more accurately and effectively deal with the data that cannot be fully observed because the observed values are limited in a certain interval, thus avoiding the loss of data information. Therefore, this paper adopts both the Fixed effect model and Tobit model for test.

significantly positive. In the following models, *RD* is R&D investment ratio, *Cr* is regional climate risk, *i* is a sample individual, *t* is time, *Control* represents the control variables, *Cash* is the cash holdings ratio, *Managshare* is the manager shareholdings ratio, α_i, μ_t and θ_k represent year-fixed, industry-fixed and province-fixed effects,[15] and $\varepsilon_{i,t}$ is the error term. We cluster standard errors at the firm-level.

$$RD_{i,t} = \beta_0 + \beta_1 Cr_{i,t-1} + \beta \sum Control_{i,t-1} + \alpha_t + \mu_j + \theta_k + \varepsilon_{i,t} \tag{4}$$

$$RD_{i,t} = \beta_0 + \beta_1 Cr_{i,t-1} + \beta_2 Cr_{i,t-1} {}^* Cash_{i,t-1} + \beta_3 Cash_{i,t-1} + \sum Control_{i,t-1} + \alpha_t + \mu_j + \theta_k + \varepsilon_{i,t} \tag{5}$$

$$RD_{i,t} = \beta_0 + \beta_1 Cr_{i,t-1} + \beta_2 Cr_{i,t-1} {}^* Managshare_{i,t-1} + \beta_3 Managshare_{i,t-1} + \sum Control_{i,t-1} + \alpha_t + \mu_j + \theta_k + \varepsilon_{i,t} \tag{6}$$

3.4. Descriptive statistics

Table 1 presents the descriptive statistical results for the main variables in this paper. It can be seen that the average of a firm's R&D investments to operating revenue ratio is 4.180 % while the standard deviation is 4.664 % (Std is close to the mean), respectively. This result shows that R&D investment intensity varies from different firms, which is consistent with the current literature (Chi et al., 2023; Vo & Le, 2017). The minimum regional climate risk (*Cr*) is only 0.168, and the maximum value is 2.530, indicating that there are great differences among the natural environment of the prefecture-level cities in China. The average cash holdings (*Cash*) is 18.2 %, the minimum value is 1.3 %, and the maximum value is 70 %, indicating that there are great differences in firms' liquidity. Average manager shareholdings is 14.0 %. In addition, variance inflation factors of all variables are less than 10, which basically excluded the influence of multicollinearity on the empirical test results of this paper.

4. Empirical results

4.1. Baseline regression results

Table 2 reports the benchmark results of the impact of the climate risk on firms' R&D investment. We show that climate risk (*Cr*) in year t-1 is significantly and negatively associated with firms' R&D investment in year t. Column (1) of Table 2 (in which $RD\_or_{i,t}$ is the dependent variable) shows that the result of the Fixed effects model, and the climate risk coefficient is significantly negative at the 1 % level. In terms of economic magnitude, the coefficient on $Cr_{i,t-1}$ of −0.190 in column (1) of Table 2 suggests that a one-standard-deviation increase in climate risk is associated with a 10.05 percentage points decrease (−0.529 × 0.190) in R&D investment to operating revenues. Column (2) of Table 2 shows that the result of the Tobit model, and the coefficient of the climate risk is still significantly negative at the level of 1 %. It can be seen that climate risk has a negative effect on firms' R&D investment, which continuously verifies H1. To increase the credibility of the above conclusions, this paper uses $RD\_asset_{i,t}$ as an alternate dependent variable to repeat the above regressions. The results are shown in column (3) and column (4) of Table 2 respectively, which are nearly the same as before. In addition, the above results are also consistent with previous studies that suggest that firms reduce R&D investment when faced with greater risk (Bloom, 2007; Julio & Yook, 2012).

4.2. The moderating effect of the cash holdings

Table 3 shows the regression results of model (5) to test H2. It can be seen from column (1) of Table 3 (in which $RD\_or_{i,t}$ is the dependent variable and the model is Fixed effects model) that the climate risk ($Cr_{i,t-1}$) has a significant negative effect on the level of firms' R&D investment at the level of 1 %. Meanwhile, the coefficient of the interaction term ($Cr_{i,t-1} \times Cash_{i,t-1}$) is significantly positive at the 1 % level, which indicates that the negative impact of climate risk on firms' R&D investment is weakened with the increase of cash holdings of firms. The regression result in column (2) of Table 3 (in which $RD\_or_{i,t}$ is the dependent variable and the Tobit model is used) is consistent with the above regression result. The regression with $RD\_or_{i,t}$ the dependent variable is consistent with the above results, which proves H2. The results show that with the increase of firms' cash holdings, firms are more able to weaken the impact of financing frictions caused by climate risk, which ultimately leads to the normal progress of firms' innovation activities. The results also show that firms' cash holdings have a significant impact on the firms' R&D investment, which is also consistent with previous findings (Brown & Petersen, 2010).

4.3. The moderating effect of manager shareholdings

Based on model (6), empirical test results of H3 are shown in Table 4. It can be seen from column (1) and (2) of Table 4 (in which $RD\_or_{i,t}$ is the dependent variable) that the coefficients of the climate risk ($Cr_{i,t-1}$) are significantly negative at the level of 1 %,

[15] Controlling the year-fixed effect can eliminate the influence of time trend and control the influence of factors such as economic environment, policy changes or social and cultural background on the regression results. Controlling industry-fixed effect can exclude the interference of industry characteristics (such as the industry's technology level, market structure, and competition degree) on the regression results. Controlling province-fixed effect can eliminate regional differences, including regional economic development level, cultural customs, natural environment and other factors.

Table 1
Descriptive statistics on the main variables.

Variable	Obs	Mean	Std.Dev.	Min	Max	VIF
$RD\_or_{i,t}$	23854	4.180	4.664	0	33.640	
$RD\_asset_{i,t}$	23854	2.071	2.071	0	12.440	
$Cr_{i,t-1}$	23854	1.196	0.529	0.168	2.530	1.05
$Cash_{i,t-1}$	23854	0.182	0.131	0.013	0.700	1.45
$Managshare_{i,t-1}$	23854	0.140	0.203	0	0.705	1.33
$Size_{i,t-1}$	23854	21.990	1.354	19.210	26.240	1.75
$Age_{i,t-1}$	23854	2.806	0.367	1.609	3.526	1.16
$Debt_{,t-1}$	23854	0.409	0.204	0.050	0.908	1.81
$Fixed_{i,t-1}$	23854	0.211	0.149	0.003	0.659	1.28
$Tbq_{i,t-1}$	23854	1.924	1.290	0	8.171	1.17
$AR_{i,t-1}$	23854	0.217	0.396	−0.302	2.486	1.23
$Fee_{i,t-1}$	23854	0.089	0.065	0	0.401	1.29
$Ceo_{i,t-1}$	23854	0.278	0.448	0	1	1.11
$BS_{i,t-1}$	23854	3.447	1.165	3	7	1.28
$Hhi_{i,t-1}$	23854	0.161	0.159	0.052	0.584	1.03
$Pgdp_{i,t-1}$	23854	9.641	4.653	3.275	19.840	1.33
$UP_{i,t-1}$	23854	1.459	1.279	0.357	5.297	1.24

Note: This table presents summary statistics and variance inflation factors of the main variables used in the empirical analyses. All variables are defined in Appendix C. All variables except *RD_or* and *RD_asset.* are in period t-1.

Table 2
Climate risk and firms' R&D investment.

	$RD\_or_{i,t}$		$RD\_asset_{i,t}$	
	Fixed effects model	Tobit model	Fixed effects model	Tobit model
	(1)	(2)	(3)	(4)
$Cr_{i,t-1}$	**−0.190******	**−0.236******	**−0.079******	**−0.100******
	(-4.392)	**(-3.202)**	**(-3.717)**	**(-2.695)**
$Size_{i,t-1}$	0.064	−0.069**	−0.117***	−0.198***
	(1.406)	(-2.224)	(-4.630)	(-12.830)
$Age_{i,t-1}$	−1.066***	−1.515***	−0.470***	−0.657***
	(-7.251)	(-15.363)	(-5.995)	(-13.301)
$Debt_{,t-1}$	−3.003***	−4.245***	−0.330**	−0.715***
	(-10.616)	(-22.999)	(-2.465)	(-7.763)
$Fixed_{i,t-1}$	−3.003***	−4.364***	−1.170***	−1.777***
	(-8.946)	(-17.368)	(-6.672)	(-14.179)
$Tbq_{i,t-1}$	0.089**	0.085***	0.072***	0.076***
	(2.216)	(3.376)	(3.439)	(5.962)
$AR_{i,t-1}$	0.684***	0.878***	0.149***	0.240***
	(8.549)	(11.019)	(4.185)	(5.990)
$Fee_{i,t-1}$	26.742***	33.498***	3.807***	6.217***
	(20.640)	(61.499)	(7.300)	(22.670)
$Ceo_{i,t-1}$	0.528***	0.656***	0.077	0.127***
	(5.610)	(9.611)	(1.542)	(3.710)
$BS_{i,t-1}$	0.031	−0.005	−0.053**	−0.079***
	(0.745)	(-0.150)	(-2.231)	(-5.088)
$Hhi_{i,t-1}$	−2.664***	−3.368***	−1.454***	−1.811***
	(-8.450)	(-15.139)	(-9.353)	(-16.255)
$Pgdp_{i,t-1}$	0.106***	0.116***	0.048***	0.052***
	(7.229)	(10.727)	(5.835)	(9.691)
$UP_{i,t-1}$	0.311***	0.358***	0.097**	0.117***
	(3.545)	(4.648)	(2.049)	(3.021)
*Cons_*	3.860***	2.651***	5.724***	4.402***
	(3.811)	(3.173)	(10.189)	(10.533)
Year FE	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.456	0.132	0.333	0.125
N	23854	23854	23854	23854

Note: This table presents the results on the relationship between climate risk and firms' R&D investment. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. All variables are defined in Appendix C.

Table 3
Climate risk, cash holdings and firms' R&D investment.

	$RD\_or_{i,t}$		$RD\_asset_{i,t}$	
	Fixed effects model	Tobit model	Fixed effects model	Tobit model
	(1)	(2)	(3)	(4)
$\boldsymbol{Cr_{i,t-1} \times Cash_{i,t-1}}$	**2.391*****	**3.075*****	**0.992*****	**1.304*****
	(7.437)	**(6.893)**	**(6.641)**	**(5.831)**
$\boldsymbol{Cr_{i,t-1}}$	**−0.576*****	**−0.753*****	**−0.237*****	**−0.316*****
	(-8.452)	**(-7.121)**	**(-7.291)**	**(-5.975)**
$Cash_{i,t-1}$	1.358***	1.736***	0.929***	1.185***
	(3.866)	(6.153)	(5.251)	(8.370)
$Size_{i,t-1}$	0.067	−0.064**	−0.115***	−0.195***
	(1.487)	(-2.085)	(-4.555)	(-12.643)
$Age_{i,t-1}$	−1.053***	−1.498***	−0.461***	−0.647***
	(-7.181)	(-15.217)	(-5.901)	(-13.114)
$Debt_{,t-1}$	−2.791***	−3.962***	−0.161	−0.492***
	(-9.361)	(-20.345)	(-1.144)	(-5.062)
$Fixed_{i,t-1}$	−2.763***	−4.037***	−0.995***	−1.538***
	(-7.898)	(-15.632)	(-5.502)	(-11.952)
$Tbq_{i,t-1}$	0.085**	0.081***	0.070***	0.073***
	(2.112)	(3.192)	(3.339)	(5.794)
$AR_{i,t-1}$	0.643***	0.828***	0.112***	0.194***
	(7.940)	(10.226)	(3.074)	(4.779)
$Fee_{i,t-1}$	26.602***	33.317***	3.731***	6.113***
	(20.621)	(61.222)	(7.195)	(22.320)
$Ceo_{i,t-1}$	0.529***	0.659***	0.076	0.128***
	(5.628)	(9.667)	(1.541)	(3.730)
$BS_{i,t-1}$	0.026	−0.012	−0.057**	−0.085***
	(0.625)	(-0.398)	(-2.411)	(-5.458)
$Hhi_{i,t-1}$	−2.650***	−3.346***	−1.447***	−1.802***
	(-8.393)	(-15.066)	(-9.315)	(-16.207)
$Pgdp_{i,t-1}$	0.104***	0.113***	0.047***	0.051***
	(7.111)	(10.483)	(5.692)	(9.370)
$UP_{i,t-1}$	0.313***	0.362***	0.096**	0.116***
	(3.567)	(4.704)	(2.033)	(3.005)
*Cons_*	3.940***	2.687***	5.622***	4.245***
	(3.883)	(3.184)	(9.959)	(10.059)
Year FE	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.458	0.133	0.336	0.126
N	23854	23854	23854	23854

Note: This table presents the results on the relationship between climate risk, cash holdings and firms' R&D investment. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. All variables are defined in Appendix C.

indicating climate risk has a significant negative impact on firms' R&D investment, which is consistent with the above results. Meanwhile, the coefficients of the interaction term ($Cr_{i,t-1} \times Managshare_{i,t-1}$) in column (1) and (2) of Table 4 are significantly positive at the level of 1 %, indicating that the negative influence of climate risk on the level of firms' R&D investment is weakened with the increase of manager shareholding ratio. The regression result in column (3) and (4) of Table 4 (in which $RD\_asset_{i,t}$ is the dependent variable) is consistent with the above result. H3 has been verified. The results also show manager shareholdings can improve manager's risk-taking and thus increases firms' R&D investment, which is consistent with other literature (Chen & Ma, 2011).

4.4. Test for robustness

4.4.1. Sample selection bias test based on heckman two-stage method

Considering that the data of firms' R&D investment are voluntarily disclosed by listed companies, it destroys the randomness of the sample, causes the sample self-selection bias, and affects the reliability of the above empirical results. To eliminate the interference of sample self-selection bias on the above empirical tests, this paper uses Heckman two-stage method for the test. In the first stage, the dependent variable is set as the dummy variable *AOE*. If firm discloses R&D investment, *AOE* is assigned 1; otherwise, *AOE* is assigned 0. The results show that the *AOE* of a total of 7201 firm-year observations are 0, accounting for about 30.2 % of the total observations, which is consistent with Chen et al. (2023). The *IMR* calculated in the first stage is then substituted into the second stage model for fitting. The models are as follows:

$$\mathrm{P}(AOE = 1) = \Phi\left(\alpha_0 + \beta \sum Control_{i,t-1} + \alpha_t + \mu_j + \theta_k + \varepsilon_{i,t}\right) \tag{7}$$

Table 4
Climate risk, management shareholdings and firms' R&D investment.

	$RD\_or_{i,t}$		$RD\_asset_{i,t}$	
	Fixed effects model	Tobit model	Fixed effects model	Tobit model
	(1)	(2)	(3)	(4)
$Cr_{i,t-1}$×$Managshare_{i,t-1}$	**0.011*****	**0.015*****	**0.007*****	**0.009*****
	(6.192)	**(5.489)**	**(7.749)**	**(6.384)**
$Cr_{i,t-1}$	**−0.326*****	**−0.460*****	**−0.164*****	**−0.227*****
	(-6.493)	**(-5.489)**	**(-6.518)**	**(-5.424)**
$Managshare_{i,t-1}$	0.806***	1.235***	0.183	0.402***
	(3.718)	(7.283)	(1.568)	(4.729)
$Size_{i,t-1}$	0.084*	−0.035	−0.115***	−0.189***
	(1.847)	(-1.105)	(-4.477)	(-12.062)
$Age_{i,t-1}$	−1.013***	−1.433***	−0.459***	−0.632***
	(-6.818)	(-14.447)	(-5.798)	(-12.710)
$Debt_{,t-1}$	−2.896***	−4.067***	−0.313**	−0.665***
	(-10.165)	(-21.816)	(-2.334)	(-7.141)
$Fixed_{i,t-1}$	−2.951***	−4.267***	−1.165***	−1.752***
	(-8.764)	(-16.967)	(-6.617)	(-13.966)
$Tbq_{i,t-1}$	0.097**	0.098***	0.073***	0.079***
	(2.422)	(3.878)	(3.472)	(6.206)
$AR_{i,t-1}$	0.642***	0.818***	0.143***	0.223***
	(7.998)	(10.191)	(4.056)	(5.535)
$Fee_{i,t-1}$	26.725***	33.498***	3.782***	6.197***
	(20.685)	(61.549)	(7.268)	(22.613)
$Ceo_{i,t-1}$	0.478***	0.579***	0.070	0.106***
	(5.067)	(8.349)	(1.388)	(3.050)
$BS_{i,t-1}$	0.030	−0.009	−0.052**	−0.079***
	(0.721)	(-0.287)	(-2.194)	(-5.109)
$Hhi_{i,t-1}$	−2.639***	−3.327***	−1.447***	−1.796***
	(-8.406)	(-14.969)	(-9.324)	(-16.132)
$Pgdp_{i,t-1}$	0.104***	0.113***	0.047***	0.051***
	(7.126)	(10.469)	(5.790)	(9.502)
$UP_{i,t-1}$	0.315***	0.363***	0.099**	0.119***
	(3.598)	(4.723)	(2.084)	(3.084)
*Cons_*	3.303***	1.828**	5.720***	4.252***
	(3.194)	(2.142)	(9.821)	(9.962)
Year FE	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
Adj_R^2/Pseudo R^2	0.458	0.133	0.335	0.126
N	23854	23854	23854	23854

Note: This table presents the results on the relationship between climate risk, manager shareholdings and firms' R&D investment. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. All variables are defined in Appendix C.

$$RD_{i,t} = \alpha_0 + \beta_1 Cr_{i,t-1} + \beta_2 IMR + \beta \sum Control_{i,t-1} + \alpha_t + \mu_j + \theta_k + \varepsilon_{i,t} \tag{8}$$

$$RD_{i,t} = \alpha_0 + \beta_1 Cr_{i,t-1} + \beta_2 Cr_{i,t-1}{*}Cash_{i,t-1} + \beta_3 Cash_{i,t-1} + \beta_4 IMR + \sum Control_{i,t-1} + \alpha_t + \mu_j + \theta_k + \varepsilon_{i,t} \tag{9}$$

$$RD_{i,t} = \alpha_0 + \beta_1 Cr_{i,t-1} + \beta_2 Cr_{i,t-1}{*}Managshare_{i,t-1} + \beta_3 Managshare_{i,t-1} + \beta_4 IMR + \sum Control_{i,t-1} + \alpha_t + \mu_j + \theta_k + \varepsilon_{i,t} \tag{10}$$

The *IMR* obtained from the first stage regression calculation was introduced into the model for re-regression, and the results are shown in Table 5. In Panel A of Table 5 (in which $RD\_or_{i,t}$ is the dependent variable), the coefficients of climate risk ($Cr_{i,t-1}$) in column (2) to column (7) are all significantly negative. The coefficients of the interaction term ($Cr_{i,t-1} \times Cash_{i,t-1}$) in column (3) and (6), and ($Cr_{i,t-1} \times Manager_{i,t-1}$) in column (4) and (7) are both significantly positive. The results in Panel B (in which $RD\_asset_{i,t}$ is the dependent variable) are also consistent with the above results. In brief, the Heckman two-stage test results of H1, H2, and H3 are completely consistent with the above empirical test results.

4.4.2. The dependent variable in period t+1 is adopted

Considering that it takes some time for firms to make and adjust R&D investment decisions, the impact of regional climate risk on firms' R&D investment may have a certain lag time. Therefore, the firms' R&D investment variable in period t+1 is used as the dependent variable for re-regression test, which not only reflects the time lag of the above impact, but also mitigates the endogenous problem in the above test. The regression results are shown in Table 6.

It can be seen from Panel A (in which $RD\_or_{i,t}$ is the dependent variable) of Table 6 that the climate risk ($Cr_{i,t-1}$) has a significant negative effect on the level of firms' R&D investment. The coefficients of the interaction term ($Cr_{i,t-1} \times Cash_{i,t-1}$) in column (2) and

Table 5
Heckman two-stage test.

	AOE	Panel A $RD\_or_{i,t}$					
		Fixed effects model			Tobit model		
	Stage-one	Stage-two					
		H1	H2	H3	H1	H2	H3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Cr_{i,t\text{-}1} \times Cash_{i,t\text{-}1}$			**2.354***** **(7.372)**			**3.014***** **(6.751)**	
$Cr_{i,t\text{-}1} \times Managshare_{i,t\text{-}1}$				**0.010***** **(6.172)**			**0.015***** **(5.456)**
$Cr_{i,t\text{-}1}$		−0.197*** (-4.532)	−0.576*** (-8.468)	−0.333*** (-6.601)	−0.247*** (-3.346)	−0.753*** (-7.119)	−0.470*** (-5.606)
$Cash_{i,t\text{-}1}$			1.357*** (3.863)			1.732*** (6.138)	
$Managshare_{i,t\text{-}1}$				0.817*** (3.764)			1.252*** (7.385)
IMR		−5.384 (-1.629)	−4.973 (-1.506)	−5.566* (-1.689)	−9.069*** (-3.832)	−8.507*** (-3.598)	−9.381*** (-3.967)
$Size_{i,t\text{-}1}$	−0.035*** (-6.272)	0.026 (0.524)	0.032 (0.649)	0.045 (0.906)	−0.131*** (-3.757)	−0.123*** (-3.525)	−0.099*** (-2.801)
$Age_{i,t\text{-}1}$	−0.096*** (-5.440)	−0.873*** (-4.556)	−0.875*** (-4.571)	−0.812*** (-4.208)	−1.189*** (-9.148)	−1.193*** (-9.189)	−1.095*** (-8.384)
$Debt_{,t\text{-}1}$	−0.217*** (-7.241)	−2.762*** (-8.964)	−2.567*** (-7.967)	−2.644*** (-8.531)	−3.844*** (-18.133)	−3.585*** (-16.216)	−3.650*** (-17.056)
$Fixed_{i,t\text{-}1}$	−0.206*** (-4.620)	−2.741*** (-7.330)	−2.520*** (-6.517)	−2.679*** (-7.147)	−3.927*** (-14.236)	−3.626*** (-12.850)	−3.813*** (-13.805)
$Tbq_{i,t\text{-}1}$	−0.007** (-2.026)	0.117*** (2.819)	0.111*** (2.668)	0.126*** (3.049)	0.132*** (4.698)	0.124*** (4.433)	0.147*** (5.211)
$AR_{i,t\text{-}1}$	0.065*** (9.191)	0.687*** (8.597)	0.646*** (7.979)	0.645*** (8.039)	0.886*** (11.109)	0.834*** (10.301)	0.824*** (10.270)
$Fee_{i,t\text{-}1}$	1.013*** (12.109)	27.661*** (19.835)	27.452*** (19.768)	27.676*** (19.908)	35.030*** (51.815)	34.756*** (51.431)	35.084*** (51.921)
$Ceo_{i,t\text{-}1}$	0.037*** (4.065)	0.523*** (5.565)	0.524*** (5.586)	0.472*** (5.010)	0.648*** (9.480)	0.651*** (9.542)	0.569*** (8.194)
$BS_{i,t\text{-}1}$	−0.007 (-1.257)	0.023 (0.558)	0.019 (0.452)	0.022 (0.527)	−0.017 (-0.536)	−0.024 (-0.760)	−0.022 (-0.691)
$Hhi_{i,t\text{-}1}$	−0.228*** (-5.896)	−2.621*** (-8.286)	−2.610*** (-8.240)	−2.595*** (-8.236)	−3.299*** (-14.786)	−3.282*** (-14.734)	−3.255*** (-14.603)
$Pgdp_{i,t\text{-}1}$	0.000* (1.750)	0.023 (0.458)	0.028 (0.541)	0.019 (0.368)	−0.023 (-0.606)	−0.017 (-0.451)	−0.031 (-0.810)
$UP_{i,t\text{-}1}$	0.006 (0.550)	−0.012 (-0.054)	0.014 (0.066)	−0.019 (-0.085)	−0.187 (-1.156)	−0.150 (-0.926)	−0.200 (-1.240)
*Cons_*	1.755*** (14.942)	7.264*** (3.167)	7.077*** (3.085)	6.810*** (2.970)	8.664*** (4.877)	8.317*** (4.682)	8.027*** (4.512)
Year FE	✓	✓	✓	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.364	0.457	0.458	0.458	0.132	0.133	0.133
N	23854	23854	23854	23854	23854	23854	23854

	AOE	Panel B $RD\_asset_{i,t}$					
		Fixed effects model			Tobit model		
	Stage-one	Stage-two					
		H1	H2	H3	H1	H2	H3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Cr_{i,t\text{-}1} \times Cash_{i,t\text{-}1}$			**0.971***** **(6.552)**			**1.273***** **(5.691)**	
$Cr_{i,t\text{-}1} \times Managshare_{i,t\text{-}1}$				**0.007***** **(7.724)**			**0.009***** **(6.342)**
$Cr_{i,t\text{-}1}$		**−0.084***** **(-3.925)**	**−0.238***** **(-7.331)**	**−0.168***** **(-6.678)**	**−0.106***** **(-2.859)**	**−0.316***** **(-5.988)**	**−0.232***** **(-5.552)**
$Cash_{i,t\text{-}1}$			0.928*** (5.249)			1.182*** (8.352)	
$Managshare_{i,t\text{-}1}$				0.189 (1.623)			0.412*** (4.844)
IMR		−3.564**	−3.418**	−3.556**	−5.401***	−5.185***	−5.455***

(*continued on next page*)

Table 5 (*continued*)

	AOE	Panel B $RD\_asset_{i,t}$					
		Fixed effects model			Tobit model		
	Stage-one	Stage-two					
		H1	H2	H3	H1	H2	H3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		(-2.415)	(-2.327)	(-2.414)	(-4.547)	(-4.371)	(-4.595)
$Size_{i,t-1}$	−0.034***	−0.142***	−0.138***	−0.139***	−0.236***	−0.231***	−0.227***
	(-6.257)	(-5.244)	(-5.141)	(-5.101)	(-13.465)	(-13.218)	(-12.827)
$Age_{i,t-1}$	−0.093***	−0.342***	−0.339***	−0.331***	−0.463***	−0.461***	−0.435***
	(-5.430)	(-3.593)	(-3.572)	(-3.446)	(-7.105)	(-7.077)	(-6.643)
$Debt_{i,t-1}$	−0.220***	−0.171	−0.008	−0.153	−0.477***	−0.263**	−0.422***
	(-7.691)	(-1.149)	(-0.051)	(-1.027)	(-4.501)	(-2.377)	(-3.948)
$Fixed_{i,t-1}$	−0.216***	−0.997***	−0.828***	−0.991***	−1.515***	−1.287***	−1.486***
	(-5.001)	(-5.227)	(-4.244)	(-5.176)	(-10.995)	(-9.134)	(-10.771)
$Tbq_{i,t-1}$	−0.008***	0.091***	0.088***	0.091***	0.103***	0.100***	0.107***
	(-2.592)	(4.172)	(4.042)	(4.204)	(7.327)	(7.100)	(7.569)
$AR_{i,t-1}$	0.066***	0.152***	0.115***	0.145***	0.244***	0.199***	0.227***
	(9.486)	(4.273)	(3.148)	(4.132)	(6.109)	(4.885)	(5.638)
$Fee_{i,t-1}$	0.998***	4.413***	4.313***	4.387***	7.121***	6.982***	7.111***
	(12.866)	(7.431)	(7.303)	(7.399)	(20.979)	(20.587)	(20.957)
$Ceo_{i,t-1}$	0.037***	0.073	0.073	0.066	0.122***	0.123***	0.100***
	(4.077)	(1.475)	(1.476)	(1.311)	(3.560)	(3.584)	(2.877)
$BS_{i,t-1}$	−0.007	−0.058**	−0.062***	−0.057**	−0.086***	−0.092***	−0.087***
	(-1.305)	(-2.440)	(-2.615)	(-2.403)	(-5.523)	(-5.874)	(-5.553)
$Hhi_{i,t-1}$	−0.234***	−1.426***	−1.420***	−1.419***	−1.770***	−1.762***	−1.754***
	(-6.222)	(-9.164)	(-9.136)	(-9.135)	(-15.838)	(-15.806)	(-15.710)
$Pgdp_{i,t-1}$	0.000*	−0.007	−0.006	−0.007	−0.030	−0.029	−0.032*
	(1.850)	(-0.290)	(-0.254)	(-0.309)	(-1.596)	(-1.519)	(-1.696)
$UP_{i,t-1}$	0.008	−0.117	−0.110	−0.115	−0.208**	−0.196**	−0.209**
	(0.779)	(-1.155)	(-1.089)	(-1.137)	(-2.562)	(-2.417)	(-2.575)
*Cons_*	1.780***	7.262***	7.074***	6.809***	8.661***	8.312***	8.024***
	(14.772)	(3.166)	(3.084)	(2.969)	(4.875)	(4.679)	(4.510)
Year FE	✓	✓	✓	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.364	0.334	0.336	0.335	0.125	0.126	0.126
N	23854	23854	23854	23854	23854	23854	23854

Note: In order to eliminate the interference of sample self-selection bias on the empirical test, this paper uses Heckman two-stage method for test. In the first stage, the dependent variable is set as the dummy variable *AOE*. If the firm discloses R&D investment, *AOE* is assigned 1; otherwise, *AOE* is assigned 0. The *IMR* calculated in the first stage is then substituted into the second stage model for fitting. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively.

column (5) from Panel A of Table 6 and ($Cr_{i,t-1} \times Manager_{i,t-1}$) in column (3) and column (6) of Table 6 are both significantly positive. The results in Panel B (in which $RD\_asset_{i,t}$ is the dependent variable) are also consistent with the above results. The above results further demonstrate the robustness of the empirical results in this paper.

4.4.3. Alternative measure of firms' R&D investment

RD_or and *RD_asset*, as the ratio of firms' R&D investment, may not include all innovation inputs, such as the number of technical staff. To ensure the robustness of the empirical test results, this paper follows Beirne et al. (2021) and uses the ratio of the number of R&D personnel in the total number of employees (*RD_*per) to measure firms' R&D investment and re-run the regression.

Table 7 reports the relevant results. Columns (1) to (6) of Table 7 present the results for H1, H2 and, H3, respectively. The coefficients on climate risk ($Cr_{i,t-1}$) in columns (1) to (6) are all significantly negative. Besides, the coefficients of the interaction term ($Cr_{i,t-1} \times Cash_{i,t-1}$) in column (2) and (5), and ($Cr_{i,t-1} \times Manager_{i,t-1}$) in column (3) and (6) are both significantly positive. H1, H2 and, H3 are all supported again.

4.5. The influence mechanism analysis

4.5.1. The influence mechanism test of firms' profitability

Based on the above analysis of the influence mechanism of climate risk affecting firms' R&D investment, climate risk will increase firms' operating cost and reduce the firms' profitability. It weakens the ability of firms to use internal profit fund to support their R&D activities, thus leading to the decline of R&D investment scale. Therefore, firms' profitability is a mediating channel through which climate risk inhibits the firms' R&D investment.

To test the mediating role of firms' profitability, this paper constructs models (11) and (12). The models are as follows:

Table 6
The dependent variable in period $t+1$ is adopted.

	Panel A $RD\_or_{i,t+1}$					
	Fixed effects model			Tobit model		
	H1	H2	H3	H1	H2	H3
	(1)	(2)	(3)	(4)	(5)	(6)
$\boldsymbol{Cr_{i,t-1} \times Cash_{i,t-1}}$		**1.076*****			**1.513*****	
		(3.081)			**(3.299)**	
$\boldsymbol{Cr_{i,t-1} \times Managshare_{i,t-1}}$			**0.004****			**0.006****
			(1.996)			**(2.262)**
$\boldsymbol{Cr_{i,t-1}}$	**−0.090***	**−0.261*****	**−0.139****	**−0.135***	**−0.384*****	**−0.225****
	(-1.813)	**(-3.574)**	**(-2.530)**	**(-1.719)**	**(-3.487)**	**(-2.575)**
$Cash_{i,t-1}$		0.815**			1.150***	
		(2.240)			(3.845)	
$Managshare_{i,t-1}$			0.837***			1.187***
			(3.673)			(6.690)
$Size_{i,t-1}$	0.020	0.022	0.046	−0.111***	−0.108***	−0.072**
	(0.436)	(0.486)	(1.006)	(-3.401)	(-3.309)	(-2.167)
$Age_{i,t-1}$	−0.752***	−0.748***	−0.696***	−1.102***	−1.097***	−1.023***
	(-5.195)	(-5.177)	(-4.771)	(-10.824)	(-10.782)	(-9.980)
$Debt_{,t-1}$	−2.538***	−2.397***	−2.413***	−3.510***	−3.303***	−3.323***
	(-8.656)	(-7.721)	(-8.175)	(-18.144)	(-16.125)	(-17.000)
$Fixed_{i,t-1}$	−3.073***	−2.919***	−3.002***	−4.357***	−4.126***	−4.241***
	(-9.219)	(-8.418)	(-8.994)	(-16.763)	(-15.408)	(-16.293)
$Tbq_{i,t-1}$	0.053	0.051	0.065	0.046	0.044	0.064**
	(1.273)	(1.235)	(1.581)	(1.636)	(1.579)	(2.277)
$AR_{i,t-1}$	0.770***	0.746***	0.719***	0.935***	0.902***	0.867***
	(8.642)	(8.143)	(8.026)	(11.073)	(10.533)	(10.191)
$Fee_{i,t-1}$	24.333***	24.263***	24.343***	30.070***	29.973***	30.096***
	(18.664)	(18.623)	(18.696)	(51.982)	(51.805)	(52.060)
$Ceo_{i,t-1}$	0.486***	0.487***	0.428***	0.594***	0.596***	0.511***
	(5.075)	(5.084)	(4.450)	(8.268)	(8.295)	(7.006)
$BS_{i,t-1}$	0.013	0.010	0.011	−0.028	−0.032	−0.033
	(0.313)	(0.253)	(0.257)	(-0.862)	(-0.988)	(-1.017)
$Hhi_{i,t-1}$	−2.662***	−2.657***	−2.635***	−3.378***	−3.369***	−3.340***
	(-8.762)	(-8.730)	(-8.702)	(-14.542)	(-14.513)	(-14.388)
$Pgdp_{i,t-1}$	0.091***	0.090***	0.089***	0.097***	0.096***	0.095***
	(5.775)	(5.702)	(5.693)	(8.513)	(8.347)	(8.306)
$UP_{i,t-1}$	0.316***	0.316***	0.318***	0.353***	0.353***	0.354***
	(3.386)	(3.381)	(3.412)	(4.207)	(4.210)	(4.227)
*Cons_*	4.016***	3.981***	3.188***	3.204***	3.100***	2.095**
	(3.953)	(3.874)	(3.038)	(3.665)	(3.504)	(2.341)
Year FE	✓	✓	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.429	0.429	0.430	0.122	0.122	0.122
N	19805	19805	19805	19805	19805	19805
	Panel B $RD\_asset_{i,t+1}$					
	Fixed effects model			Tobit model		
	H1	H2	H3	H1	H2	H3
	(1)	(2)	(3)	(4)	(5)	(6)
$\boldsymbol{Cr_{i,t-1} \times Cash_{i,t-1}}$		**0.420*****			**0.630*****	
		(2.637)			**(2.747)**	
$\boldsymbol{Cr_{i,t-1} \times Managshare_{i,t-1}}$			**0.003*****			**0.004*****
			(2.900)			**(2.720)**
$\boldsymbol{Cr_{i,t-1}}$	**−0.051****	**−0.114*****	**−0.084*****	**−0.079****	**−0.180*****	**−0.133*****
	(-2.163)	**(-3.271)**	**(-3.155)**	**(-2.025)**	**(-3.263)**	**(-3.036)**
$Cash_{i,t-1}$		0.797***			1.033***	
		(4.384)			(6.898)	
$Managshare_{i,t-1}$			0.199*			0.380***
			(1.660)			(4.276)
$Size_{i,t-1}$	−0.120***	−0.117***	−0.114***	−0.197***	−0.194***	−0.185***
	(-4.860)	(-4.792)	(-4.563)	(-12.098)	(-11.928)	(-11.195)
$Age_{i,t-1}$	−0.356***	−0.352***	−0.343***	−0.507***	−0.503***	−0.482***
	(-4.572)	(-4.532)	(-4.372)	(-9.946)	(-9.879)	(-9.397)
$Debt_{,t-1}$	−0.315**	−0.153	−0.289**	−0.637***	−0.422***	−0.581***
	(-2.291)	(-1.054)	(-2.085)	(-6.598)	(-4.129)	(-5.952)

(*continued on next page*)

Table 6 (*continued*)

	Panel B $RD\_asset_{i,t+1}$					
	Fixed effects model			Tobit model		
	H1	H2	H3	H1	H2	H3
	(1)	(2)	(3)	(4)	(5)	(6)
$Fixed_{i,t-1}$	−1.121***	−0.961***	−1.107***	−1.681***	−1.462***	−1.647***
	(-6.408)	(-5.319)	(-6.309)	(-12.984)	(-10.973)	(-12.699)
$Tbq_{i,t-1}$	0.021	0.020	0.023	0.018	0.018	0.024*
	(1.051)	(1.021)	(1.175)	(1.321)	(1.281)	(1.702)
$AR_{i,t-1}$	0.177***	0.144***	0.166***	0.248***	0.207***	0.228***
	(4.482)	(3.584)	(4.252)	(5.856)	(4.827)	(5.336)
$Fee_{i,t-1}$	4.290***	4.238***	4.286***	6.461***	6.389***	6.462***
	(8.153)	(8.072)	(8.155)	(22.226)	(21.992)	(22.233)
$Ceo_{i,t-1}$	0.069	0.068	0.057	0.111***	0.110***	0.086**
	(1.390)	(1.374)	(1.132)	(3.082)	(3.070)	(2.365)
$BS_{i,t-1}$	−0.047**	−0.050**	−0.047**	−0.071***	−0.075***	−0.072***
	(-1.999)	(-2.143)	(-1.998)	(-4.419)	(-4.691)	(-4.485)
$Hhi_{i,t-1}$	−1.461***	−1.459***	−1.454***	−1.827***	−1.824***	−1.814***
	(-9.654)	(-9.639)	(-9.630)	(-15.720)	(-15.721)	(-15.609)
$Pgdp_{i,t-1}$	0.044***	0.043***	0.044***	0.048***	0.046***	0.047***
	(5.263)	(5.136)	(5.237)	(8.322)	(8.047)	(8.187)
$UP_{i,t-1}$	0.088*	0.086*	0.088*	0.102**	0.100**	0.102**
	(1.821)	(1.785)	(1.828)	(2.429)	(2.377)	(2.437)
*Cons_*	5.449***	5.263***	5.300***	4.447***	4.187***	4.146***
	(9.890)	(9.502)	(9.221)	(10.178)	(9.479)	(9.262)
Year FE	✓	✓	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.319	0.321	0.319	0.118	0.119	0.120
N	19805	19805	19805	19805	19805	19805

Note: To exclude the endogenous interference of the empirical test in this paper, the above regression was repeated with the firms' R&D investment in t+1 period. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. All variables are defined in Appendix C.

$$Roa_{i,t} = \beta_0 + \beta_1 Cr_{i,t-1} + \beta \sum Control_{i,t-1} + \gamma_t + \delta_j + \theta_k + \varepsilon_{i,t} \tag{11}$$

$$RD_{i,t} = \beta_0 + \beta_1 Cr_{i,t-1} + \beta_2 Roa_{i,t} + \beta \sum Control_{i,t-1} + \gamma_t + \delta_j + \theta_k + \varepsilon_{i,t} \tag{12}$$

Refer to the existing literature (Guan et al., 2021), this paper uses the return on assets (*Roa*) as the proxy variable of firms' profitability. The regression results are shown in Table 8. It can be seen from columns (1) of Table 8 (in which *Roa* is the dependent variable) that the climate risk (*Cr*) has a significant negative effect on the firms' profitability (*Roa*) at the level of 10 %, which indicates that climate risk leads to a decrease in firms' profitability. The Columns (2) and (3) of Table 8 list the regression results with the introduction of firms' profitability (*Roa*) (in which $RD\_or_{i,t}$ is the dependent variable). After the introduction of firms' profitability (*Roa*), the regression coefficients of climate risk are still significantly negative at least 5 % level, but its absolute value decreases compared to the regression coefficient before the introduction of firms' profitability (*Roa*) (in columns (1) and (2) of Table 2), and the regression coefficients of firms' profitability (*Roa*) are significantly positive at the level of 1 %, indicating that climate risk partly reduces the firms' R&D investment by reducing their profitability. Columns (4) and (5) of Table 8 show the regression test results based on the Tobit model, and the results remain consistent.

4.5.2. The influence mechanism test of firms' financing constraint

Based on the analytical logic of H1 above, climate risk affects firms' R&D investment partly by increasing firms' financing costs and exacerbating the external financing constraint faced by firms, making it difficult for firms to provide financial support through external financing, that is, financing constraint also plays a mediating role in the impact of climate risk on firms' R&D investment. In order to test the mediating role of firms' financing constraint, this paper constructs models (13) and (14). The models are as follows:

$$Cost_{i,t-1} = \beta_0 + \beta_1 Cr_{i,t-1} + \beta \sum Control_{i,t-1} + \gamma_t + \delta_j + \theta_k + \varepsilon_{i,t} \tag{13}$$

$$RD_{i,t} = \beta_0 + \beta_1 Cr_{i,t-1} + \beta_2 Cost_{i,t-1} + \beta \sum Control_{i,t-1} + \gamma_t + \delta_j + \theta_k + \varepsilon_{i,t} \tag{14}$$

Refers to the existing literature (Zheng & Shen, 2024), this paper uses the ratio of interest expense to average of long-term debt and short-term debt (*Cost*) to measure firms' external financing constraint.

The regression results are shown in Table 9. It can be seen from columns (1) of Table 9 (in which *Cost* is the dependent variable) that the climate risk has a significant positive effect on the firms' financing constraint (*Cost*) at the level of 1 %, which indicating that climate risk will increase the financing cost of the firm and exacerbate their financing constraint. The coefficients of the firms'

Table 7
Alternative measure of firms' R&D investment.

	$RD\_per_{i,t}$					
	Fixed effects model			Tobit model		
	H1	H2	H3	H1	H2	H3
	(1)	(2)	(3)	(4)	(5)	(6)
$Cr_{i,t-1} \times Cash_{i,t-1}$		**3.842*** **			**2.906** **	
		(3.703)			**(2.109)**	
$Cr_{i,t-1} \times Managshare_{i,t-1}$			**0.022*** **			**0.010*** **
			(3.798)			**(3.809)**
$Cr_{i,t-1}$	**−0.470*** **	**−1.118*** **	**−0.751*** **	**−0.395* **	**−0.835*** **	**−0.281*** **
	(-3.401)	**(-5.648)**	**(-4.813)**	**(-1.838)**	**(-2.788)**	**(-4.180)**
$Cash_{i,t-1}$		3.108***			0.278	
		(-3.153)			(0.338)	
$Managshare_{i,t-1}$			2.297***			0.192
			(3.632)			(-1.254)
$Size_{i,t-1}$	−0.062	−0.075	−0.001	0.096	0.094	0.563***
	(-0.520)	(-0.628)	(-0.010)	(1.144)	(1.115)	(22.192)
$Age_{i,t-1}$	−1.336***	−1.365***	−1.182***	−2.420***	−2.417***	0.047
	(-3.533)	(-3.606)	(-3.101)	(-8.346)	(-8.335)	(0.558)
$Debt_{,t-1}$	−2.574***	−3.402***	−2.253***	−5.733***	−5.742***	0.011
	(-3.658)	(-4.530)	(-3.169)	(-11.153)	(-10.759)	(0.070)
$Fixed_{i,t-1}$	−12.340***	−13.057***	−12.177***	−18.844***	−18.807***	−2.346***
	(-14.120)	(-14.182)	(-13.886)	(-27.276)	(-26.626)	(-11.501)
$Tbq_{i,t-1}$	0.810***	0.807***	0.834***	0.911***	0.908***	0.154***
	(7.843)	(7.855)	(8.124)	(13.193)	(13.139)	(7.395)
$AR_{i,t-1}$	0.958***	1.178***	0.833***	1.809***	1.814***	−0.024
	(4.501)	(5.382)	(3.927)	(7.668)	(7.608)	(-0.342)
$Fee_{i,t-1}$	15.828***	15.858***	15.819***	27.660***	27.554***	9.060***
	(6.570)	(6.570)	(6.580)	(18.175)	(18.100)	(21.661)
$Ceo_{i,t-1}$	1.015***	1.025***	0.865***	1.448***	1.449***	0.084
	(3.968)	(4.009)	(3.338)	(7.599)	(7.608)	(1.442)
$BS_{i,t-1}$	0.513***	0.533***	0.508***	0.876***	0.877***	0.094***
	(4.986)	(5.159)	(4.966)	(10.139)	(10.140)	(3.783)
$Hhi_{i,t-1}$	−4.752***	−4.744***	−4.684***	−6.205***	−6.196***	−1.678***
	(-5.100)	(-5.089)	(-5.035)	(-10.043)	(-10.029)	(-9.069)
$Pgdp_{i,t-1}$	0.279***	0.283***	0.274***	0.276***	0.276***	0.003
	(7.315)	(7.446)	(7.219)	(9.544)	(9.544)	(0.334)
$UP_{i,t-1}$	1.366***	1.396***	1.376***	1.033***	1.036***	0.217***
	(5.829)	(5.974)	(5.875)	(5.100)	(5.114)	(3.466)
*Cons_*	11.095***	13.171***	9.267***	−19.955***	−19.410***	−14.214***
	(4.097)	(4.832)	(3.319)	(-8.412)	(-8.078)	(-20.232)
Year FE	✓	✓	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓	✓
Adj_R^2/Pseudo R^2	0.495	0.496	0.496	0.177	0.177	0.085
N	23854	23854	23854	23854	23854	23854

Note: This paper follows Beirne et al. (2021) and uses the ratio of the number of R&D personnel in the total number of employees to measure firms' R&D investment. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. All variables are defined in Appendix C.

financing constraint (*Cost*) from columns (2) to (5) of Table 9 are significantly negative. At the same time, after the introduction of firms' financing constraint (*Cost*), the regression coefficients of climate risk (*Cr*) are still significantly negative from columns (2) to (5), but compared with the regression coefficient without the addition of firms' financing constraint (*Cost*) (in columns (1) to (4) of Table 2), its absolute value decreases, indicating that firms' financing constraint plays a mediating effect in the negative effect of climate risk on firms' R&D investment, that is, climate risk reduces the firms' R&D investment partly by increasing firms' financing costs and exacerbating the external financing faced by firms.

Table 8
The influence mechanism test of firms' profitability.

	$Roa_{i,t}$	$RD\_or_{i,t}$		$RD\_asset_{i,t}$	
		Fixed effects model	Tobit model	Fixed effects model	Tobit model
	(1)	(2)	(3)	(4)	(5)
$Cr_{i,t-1}$	**−0.002***	**−0.181*****	**−0.222*****	**−0.069*****	**−0.084****
	(-1.762)	**(-4.288)**	**(-3.150)**	**(-3.419)**	**(-2.574)**
$Roa_{i,t}$		**2.809*****	**3.236*****	**4.247*****	**4.772*****
		(4.715)	**(6.626)**	**(14.459)**	**(19.631)**
$Size_{i,t-1}$	0.003***	0.054	−0.079**	−0.132***	−0.213***
	(5.374)	(1.197)	(-2.554)	(-5.341)	(-13.906)
$Age_{i,t-1}$	−0.006***	−1.050***	−1.498***	−0.446***	−0.633***
	(-2.808)	(-7.165)	(-15.214)	(-5.796)	(-12.957)
$Debt_{,t-1}$	−0.090***	−2.749***	−3.958***	0.054	−0.288***
	(-23.214)	(-9.550)	(-20.913)	(0.403)	(-3.084)
$Fixed_{i,t-1}$	−0.011**	−2.971***	−4.327***	−1.122***	−1.715***
	(-2.352)	(-8.873)	(-17.239)	(-6.514)	(-13.854)
$Tbq_{i,t-1}$	0.006***	0.072*	0.066***	0.046**	0.047***
	(10.280)	(1.782)	(2.585)	(2.257)	(3.703)
$AR_{i,t-1}$	0.016***	0.638***	0.827***	0.080**	0.163***
	(13.270)	(7.948)	(10.332)	(2.307)	(4.089)
$Fee_{i,t-1}$	−0.115***	27.065***	33.862***	4.296***	6.737***
	(-9.064)	(20.782)	(61.916)	(8.343)	(24.737)
$Ceo_{i,t-1}$	−0.003**	0.536***	0.666***	0.089*	0.140***
	(-2.167)	(5.714)	(9.762)	(1.831)	(4.142)
$BS_{i,t-1}$	−0.005***	0.045	0.014	−0.032	−0.051***
	(-8.214)	(1.085)	(0.454)	(-1.370)	(-3.336)
$Hhi_{i,t-1}$	−0.008*	−2.642***	−3.341***	−1.421***	−1.770***
	(-1.922)	(-8.414)	(-15.038)	(-9.247)	(-16.087)
$Pgdp_{i,t-1}$	0.000	0.105***	0.115***	0.047***	0.051***
	(0.830)	(7.229)	(10.677)	(5.886)	(9.620)
$UP_{i,t-1}$	−0.002	0.318***	0.365***	0.107**	0.128***
	(-1.644)	(3.622)	(4.749)	(2.297)	(3.348)
*Cons_*	0.039**	3.751***	2.585***	5.559***	4.298***
	(2.565)	(3.702)	(3.098)	(10.096)	(10.411)
Year FE	✓	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓
Adj_R^2/Pseudo R^2	0.135	0.458	0.133	0.349	0.129
N	23854	23854	23854	23854	23854

Note: This paper examines the influence mechanism of climate risk inhibiting firms' R&D investment by reducing firms' profitability. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. The firms' profitability (*Roa*) is return on assets. The main variables are defined in Appendix C.

4.6. Further analysis based on China's institutional environment

4.6.1. Further analysis based on the property right of firms

In China's capital market, although the number of listed private enterprises and foreign-funded enterprises has gradually increased, and the proportion of listed state-owned firms (SOEs) has decreased, the market value of SOEs dominates the capital market.[16] As the controlling shareholders of SOEs, the central government and local governments still have a unique "Paternalism". One of the manifestations is to provide financial support for SOEs (Brandt & Li, 2003). Since SOEs often bear policy burdens such as ensuring employment and local economic development (Fan et al., 2007), local governments have stronger incentives to provide government subsidies and take other protective measures. The Chinese government provides special subsidies for SOEs such as technological innovation subsidies, industrial upgrading subsidies and environmental protection renovation subsidies, reduces the income tax rate of small-scale SOEs, and provides tax exemptions, tax rebates and tax credits for SOEs in specific industries.[17] In addition, manager of SOEs is usually appointed by various levels of government or State-owned Assets Supervision and Administration Commission (SASAC). Therefore, they are like quasi-government officials and have close political ties with the government (Fan et al., 2007). Relying on close political relations, SOEs are easy to get government subsidies. In the meantime, in the credit market, due to the dominant position of state-owned commercial banks, which have the same controlling shareholders as SOEs, as well as the monopoly

[16] According to *Annual Statistical Report on the Development of Chinese Listed Compani*es (2022) released by China Association of Public Companies published, it can be seen that by the end of 2022, there were 1351 state-owned listed enterprises, accounting for 26.6 percent of the number of listed companies in China. The market capitalization of state-owned listed enterprises accounts for 45 percent of the total market capitalization of listed enterprises in China.

[17] See more details for https://www.most.gov.cn/index.html.

Table 9
The influence mechanism test of firms' financing constraint.

	$Cost_{i,t}$	$RD\_or_{i,t}$		$RD\_asset_{i,t}$	
		Fixed effects model	Tobit model	Fixed effects model	Tobit model
	(1)	(2)	(3)	(4)	(5)
$Cr_{i,t-1}$	**0.003*****	**−0.183*****	**−0.228*****	**−0.072*****	**−0.091*****
	(4.339)	**(-4.221)**	**(-3.086)**	**(-3.657)**	**(-2.649)**
$Cost_{i,t}$		**−2.393*****	**−2.743*****	**−0.423***	**−0.538***
		(-3.606)	**(-4.475)**	**(-1.833)**	**(-1.751)**
$Size_{i,t-1}$	−0.001**	0.061	−0.071**	−0.118***	−0.199***
	(-2.151)	(1.351)	(-2.288)	(-4.648)	(-12.855)
$Age_{i,t-1}$	0.003**	−1.058***	−1.504***	−0.468***	−0.655***
	(2.479)	(-7.200)	(-15.258)	(-5.973)	(-13.258)
$Debt_{,t-1}$	0.041***	−2.906***	−4.130***	−0.313**	−0.693***
	(12.098)	(-10.330)	(-22.173)	(-2.335)	(-7.448)
$Fixed_{i,t-1}$	0.008**	−2.983***	−4.332***	−1.167***	−1.771***
	(2.202)	(-8.907)	(-17.242)	(-6.649)	(-14.126)
$Tbq_{i,t-1}$	0.001	0.091**	0.087***	0.073***	0.076***
	(1.444)	(2.260)	(3.440)	(3.453)	(5.987)
$AR_{i,t-1}$	−0.004***	0.673***	0.869***	0.147***	0.238***
	(-4.058)	(8.405)	(10.901)	(4.140)	(5.940)
$Fee_{i,t-1}$	0.008	26.760***	33.521***	3.810***	6.221***
	(0.803)	(20.733)	(61.567)	(7.315)	(22.686)
$Ceo_{i,t-1}$	−0.001	0.526***	0.654***	0.076	0.127***
	(-0.737)	(5.597)	(9.584)	(1.536)	(3.697)
$BS_{i,t-1}$	−0.000	0.030	−0.005	−0.053**	−0.079***
	(-0.232)	(0.741)	(-0.172)	(-2.233)	(-5.097)
$Hhi_{i,t-1}$	0.001	−2.660***	−3.367***	−1.453***	−1.811***
	(0.394)	(-8.470)	(-15.143)	(-9.358)	(-16.254)
$Pgdp_{i,t-1}$	−0.000**	0.105***	0.115***	0.048***	0.052***
	(-2.405)	(7.182)	(10.633)	(5.819)	(9.651)
$UP_{i,t-1}$	0.002	0.315***	0.363***	0.098**	0.118***
	(1.540)	(3.589)	(4.712)	(2.064)	(3.046)
*Cons_*	0.043***	3.963***	2.799***	5.742***	4.431***
	(3.976)	(3.911)	(3.349)	(10.228)	(10.596)
Year FE	✓	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.047	0.457	0.132	0.334	0.125
N	23854	23854	23854	23854	23854

Note: This paper examines the influence mechanism of climate risk affects firms' R&D investment partly by increasing firms' financing costs and exacerbating the external financing faced by firms. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. The firms' financing constraint (*Cost*) is the ratio of interest expense to the average of long-term debt and short-term debt. The main variables are defined in Appendix C.

position of state-owned firms and the implicit guarantee of the government, state-owned commercial banks prefer to issue credit funds to state-owned firms. Therefore, SOEs are more likely to receive direct financial support from the government and credit of commercial banks, making them maintain the level of R&D expenditure when facing climate risk. Thus, it can be expected that compared with state-owned firms' R&D investment, climate risk has a more significant negative impact on non-state-owned firms' R&D investment.

This paper introduces the interaction term of the climate risk and the property right of firms ($Cr_{i,t-1} \times SOE_{i,t-1}$) to study the property right heterogeneity. Property right (*SOE*) is a dummy variable. When the firm is a SOEs, *SOE* is assigned the value of 1; otherwise, *SOE* is assigned the value of 0. Descriptive statistics show that the sample size of state-owned listed companies is 7330, accounting for about 30.7 % of the total number of samples. The regression results are shown in Table 10. It can be seen from column (1) and (2) of Table 10 (in which $RD\_or_{i,t}$ is the dependent variable) that the climate risk ($Cr_{i,t-1}$) has significant negative effects on the level of firms' R&D investment at the level of 1 %. Besides, the coefficient of the interaction term ($Cr_{i,t-1} \times SOE_{i,t-1}$) is significantly positive at the level of 1 %, indicating that if the nature of firms' property right is state-owned, the negative influence of climate risk on the level of firms' R&D investment is weakened, which is consistent with the above expectation. To ensure the robustness of the above result, this paper uses $RD\_asset_{i,t}$ as alternative dependent variable. The result is shown respectively in column (3) and (4) of Table 10 and remains nearly the same as before.

4.6.2. Further analysis based on government subsidies

Government subsidies can provide financial support for firms, increase the innovation ability of firms and promote the level of R&D investment (Xu et al., 2023). In order to promote firms' R&D investment, the Chinese government often provides funds for firms through government subsidies to facilitate firms' innovation. The Chinese government mainly provides government subsidies in three ways: financial allocation, tax preference and financial discount interest. For example, when a firm is identified as a high-technology

Table 10
Further analysis based on the property right of firms.

	$RD\_or_{i,t}$		$RD\_asset_{i,t}$	
	Fixed effects model	Tobit model	Fixed effects model	Tobit model
	(1)	(2)	(3)	(4)
$Cr_{i,t-1} \times Soe_{i,t-1}$	**0.673*** **	**0.605*** **	**0.259*** **	**0.220*** **
	(8.911)	**(4.875)**	**(7.076)**	**(3.547)**
$Cr_{i,t-1}$	**−0.433*** **	**−0.421*** **	**−0.173*** **	**−0.167*** **
	(-7.352)	**(-5.048)**	**(-6.108)**	**(-3.998)**
$Soe_{i,t-1}$	−0.461***	−0.775***	−0.148**	−0.289***
	(-3.971)	(-9.205)	(-2.280)	(-6.847)
$Size_{i,t-1}$	0.099**	−0.017	−0.105***	−0.179***
	(2.194)	(-0.549)	(-4.110)	(-11.471)
$Age_{i,t-1}$	−0.984***	−1.392***	−0.443***	−0.611***
	(-6.629)	(-14.052)	(-5.610)	(-12.287)
$Debt_{,t-1}$	−2.897***	−4.053***	−0.296**	−0.643***
	(-10.192)	(-21.880)	(-2.199)	(-6.945)
$Fixed_{i,t-1}$	−2.881***	−4.183***	−1.129***	−1.711***
	(-8.581)	(-16.652)	(-6.433)	(-13.642)
$Tbq_{i,t-1}$	0.094**	0.089***	0.074***	0.077***
	(2.323)	(3.516)	(3.519)	(6.068)
$AR_{i,t-1}$	0.640***	0.812***	0.135***	0.215***
	(8.051)	(10.181)	(3.792)	(5.360)
$Fee_{i,t-1}$	26.686***	33.353***	3.790***	6.163***
	(20.719)	(61.388)	(7.302)	(22.495)
$Ceo_{i,t-1}$	0.431***	0.507***	0.045	0.072**
	(4.530)	(7.279)	(0.877)	(2.045)
$BS_{i,t-1}$	0.078*	0.076**	−0.038	−0.049***
	(1.843)	(2.364)	(-1.532)	(-3.086)
$Hhi_{i,t-1}$	−2.659***	−3.362***	−1.452***	−1.810***
	(-8.590)	(-15.156)	(-9.446)	(-16.262)
$Pgdp_{i,t-1}$	0.107***	0.117***	0.048***	0.053***
	(7.306)	(10.820)	(5.872)	(9.748)
$UP_{i,t-1}$	0.324***	0.378***	0.101**	0.124***
	(3.692)	(4.925)	(2.138)	(3.225)
*Cons_*	0.099**	−0.017	−0.105***	−0.179***
	(2.194)	(-0.549)	(-4.110)	(-11.471)
Year FE	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.459	0.133	0.335	0.126
N	23854	23854	23854	23854

Note: This table presents the results on the further analysis based on the property right of firms. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. Property right ($SOE_{i,t-1}$) is a dummy variable. When the firm is a state-owned enterprise, *SOE* is 1; otherwise, *SOE* is 0.

firm, it can apply for financial subsidies from the local government. The Chinese government has set up a special innovation fund, which is open only to science and technology innovation firms. With the help of various types of government subsidies, the negative effect of climate risk on the firms' investment capacity will be weakened. Therefore, it can be expected that comparing with firms with higher government subsidies, the negative impact of climate risk on the level of firms' R&D investment is more significant in firms with lower government subsidies. Considering the availability of data and the consistency of data caliber, this paper selects the logarithm of total government subsidies +1 to measure government subsidies (*Sub*) by referring to existing literature (Jiang & Jiang, 2024; Zhang et al., 2024). Based on Model (4), the government subsidies to total assets ($Sub_{i,t-1}$) and the interaction term of climate risk and government subsidies to total assets ($Cr_{i,t-1} \times Sub_{i,t-1}$) are introduced to study the effect of government subsidies.

Table 11 shows the relevant regression results. It can be seen from column (1) and (2) of Table 11 (in which $RD\_or_{i,t}$ is the dependent variable) that the climate risk ($Cr_{i,t-1}$) has significant negative effects on the level of firms' R&D investment at the level of 1 %. Meanwhile, the coefficient of the interaction term ($Cr_{i,t-1} \times Sub_{i,t-1}$) is significantly positive at the level of 1 %, which indicates that the negative influence of climate risk on the level of firms' R&D investment is weakened with the increase of government subsidies. The results in column (3) and (4) (in which $RD\_asset_{i,t}$ is the dependent variable) are consistent with the above result.

5. Conclusion

With the continuous deterioration of the global climate, climate risk caused by climate change has a significant impact on macroeconomic development and micro firms' operation activities. This paper designs a measurement indicator of regional climate risk and investigates the impact of regional climate risk on firms' R&D investment based on the data of 23854 firm-year in China from 2008 to 2021. Our findings show that regional climate risk has a negative effect on firms' R&D investment level. We further find that

Table 11
Further analysis based on government subsidies.

	$RD\_or_{i,t}$		$RD\_asset_{i,t}$	
	Fixed effects model	Tobit model	Fixed effects model	Tobit model
	(1)	(2)	(3)	(4)
$Cr_{i,t-1} \times Sub_{i,t-1}$	**0.040*****	**0.049*****	**0.019*****	**0.023*****
	(2.657)	**(3.036)**	**(2.832)**	**(2.870)**
$Cr_{i,t-1}$	**−0.811*****	**−0.998*****	**−0.381*****	**−0.459*****
	(-3.336)	**(-3.813)**	**(-3.414)**	**(-3.512)**
$Sub_{i,t-1}$	0.034	0.082***	0.007	0.014
	(1.265)	(3.891)	(-0.515)	(1.381)
$Size_{i,t-1}$	−0.043	−0.259***	−0.138***	−0.255***
	(-0.833)	(-7.499)	(-4.922)	(-14.726)
$Age_{i,t-1}$	−1.068***	−1.517***	−0.470***	−0.658***
	(-7.306)	(-15.431)	(-6.006)	(-13.326)
$Debt_{,t-1}$	−2.880***	−4.004***	−0.306**	−0.644***
	(-10.345)	(-21.645)	(-2.289)	(-6.953)
$Fixed_{i,t-1}$	−2.957***	−4.307***	−1.162***	−1.760***
	(-8.820)	(-17.176)	(-6.618)	(-14.048)
$Tbq_{i,t-1}$	0.065	0.040	0.069***	0.063***
	(1.583)	(1.574)	(3.240)	(4.890)
$AR_{i,t-1}$	0.670***	0.850***	0.146***	0.232***
	(8.397)	(10.692)	(4.121)	(5.792)
$Fee_{i,t-1}$	26.553***	33.158***	3.779***	6.120***
	(20.563)	(60.978)	(7.245)	(22.296)
$Ceo_{i,t-1}$	0.473***	0.554***	0.067	0.098***
	(5.030)	(8.074)	(1.347)	(2.834)
$BS_{i,t-1}$	−0.030	−0.131***	−0.064***	−0.116***
	(-0.698)	(-3.982)	(-2.581)	(-7.043)
$Hhi_{i,t-1}$	−2.645***	−3.321***	−1.450***	−1.798***
	(-8.477)	(-14.976)	(-9.366)	(-16.153)
$Pgdp_{i,t-1}$	0.105***	0.114***	0.048***	0.052***
	(7.221)	(10.578)	(5.822)	(9.591)
$UP_{i,t-1}$	0.290***	0.321***	0.093**	0.106***
	(3.316)	(4.180)	(1.968)	(2.745)
*Cons_*	5.942***	8.302***	6.329***	6.241***
	(5.522)	(8.519)	(10.624)	(12.803)
Year FE	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
Adj_R^2/Pseudo R^2	0.459	0.133	0.334	0.126
N	23854	23854	23854	23854

Note: This table presents the results on the further analysis based on the government subsidies. T-value is reported in parentheses. *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. Government subsidies ($Sub_{i,t-1}$) is measured by the Ln(Total government subsidies+1).

abovementioned impact will be weakened with the increase of firms' cash holdings level and manager shareholdings level. The relevant mechanism analysis shows that climate risk reduces firms' R&D investment by decreasing firms' profitability and exacerbating firms' external financing constraint. Further analysis based on China's institutional environment also shows that the above impact of climate risk on firms' R&D investment is more pronounced for non-SOEs and firms with lower government subsidies.

In the era of increasingly fierce global competition, firms' R&D innovation is not only the key driver to enhance international competitiveness and achieve sustainable development, but also an important path for firms to achieve high-tech, green and low-carbon development. With the intensification of global climate change and the frequent occurrence of extreme weather events, the impact of climate risk on firms' R&D innovation should be taken seriously. The governments and firms need to work together to take effective measures to address the challenges posed by climate risk.

At the firm level, first, firms should adopt reasonable cash management strategies and dynamically adjust their cash reserves to ensure that they have sufficient cash reserves to support the R&D activities when extreme weather events and other emergencies occur. Secondly, firms should strengthen the disclosure of climate risk related information and establish a sound climate risk information disclosure system. The above measures can help investors and stakeholders better understand the climate risk faced by firms, thus reducing the degree of information asymmetry between firms and investors and effectively alleviating the exacerbation of climate risk on firms' financing constraints. Third, when firms face the climate risk, shareholders can adopt effective corporate governance measures, including equity incentives, to motivate the manager to pay attention to the long-term development of firms, and alleviate the negative effect of climate risk on corporate innovation activities.

At the government level, first of all, the government should establish and improve the climate risk assessment and early warning system, provide timely and accurate climate risk information for firms, which help firms make arrangements in advance and respond effectively. Secondly, the government should further promote market-oriented reform and reduce administrative intervention, which will help non-SOEs obtain more resources through market competition and enhance independent innovation and sustainable development capabilities. Finally, government subsidies and other policies should be introduced to alleviate the negative effect of climate risk on firms' R&D investment capacity and encourage them to carry out innovative activities, especially for non-SOEs and other firms with high degree of financing constraints.

CRediT authorship contribution statement

Yong Wang: Conceptualization, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Chao Wang:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.

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Appendix A

List of 251 prefecture-level cities.

This table lists the 251 prefecture-level cities where the research samples are located and the corresponding number of firm-year observations in each prefecture-level city, the number of firm-year observations of the region is in parentheses to the right of the region.

Province/Municipality	City
Anhui(852)	Wuhu(114), Hefei(325), Bozhou(41), Tongling(81), Anqing(26), Xuancheng(33), Bengbu(46), Huangshan(23), Huaibei (33), Ma on Shan(79), Chuzhou(41), Huainan(7), Fuyang(12), Luan(20).
Beijing(2424)	Beijing(2424).
Chongqing(291)	Chongqing(291).
Fujian(846)	Xiamen(327), Fuzhou(260), Sanming(13), Zhangzhou(33), Quanzhou(69), Putian(13), Nanping(73), Longyan(47), Ningde (11).
Gansu(236)	Baiyin(18), Lanzhou(137), Jiayuguan(19), Tianshui(18), Longnan(17), Jiuquan(25).
Guangdong(4290)	Shenzhen (2010), Huizhou(84), Foshan(318), Guangzhou(733), Zhuhai(203), Shantou(243), Dongguan(194), Jiangmen (69), Shaoguan(38), Zhaoqing(71), Maoming(8), Zhongshan(160), Meizhou(47), Jieyang(28), Chaozhou(43), Zhanjiang (23), Yunfu(10), Yangjiang(5), Qingyuan(3).
Guangxi Zhuang Autonomous Region(257)	Liuzhou(54), Beihai(33), Yulin(12), Guigang(9), Nanning(66), Guilin(51), Wuzhou(21), Hezhou(9), Hechi(2).
Guizhou(212)	Guiyang(162), Qiannan Buyi and Miao Autonomous Prefecture(10), Anshun(11), Zunyi(18), Liupanshui(10), Tongren(1).
Hainan(174)	Haikou(164), Sanya(10).
Hebei(462)	Shijiazhuang(153), Tangshan(90), Handan(12), Chengde(13), Zhangjiakou(11), Xingtai(17), Cangzhou(45). Langfang(11), Baoding(79), Qinhuangdao(15), Hengshui(14).
Heilongjiang(293)	Harbin(202), Jiamusi(10), Daqing(13), Qiqihar(24), Mudan River(22), Qitai River(10), Jixi(6), Yichun(6).
Henan(652)	Xuchang(47), Zhengzhou(215), Jiaozuo(62), Luohe(25), Xinxiang(28), Luoyang(95), Nanyang(44), Puyang(18), Kaifeng (11), Anyang(24), Xinyang(24), Pingdingshan(40), Sanmenxia(2), Zhoukou(9), Shangqiu(5), Zhumadian(2), Hebi(1).
Hubei(728)	Yichang(57), Jingzhou(61), Xiangyang(84), Jingmen(35), Wuhan(402), Xiaogan, Huangshi, Ezhou(11), Xianning(4), Shiyan (20), Huangshan(49), Suizhou(5).
Hunan(733)	Changsha(403), Xiangtan(40), Hengyang(27), Yueyang(75), Zhuzhou(87), Yongzhou(18), Huaihua(12), Yiyang(33), Chenzhou(14), Changde(24).
Inner Mongolia Autonomous Region(144)	Ordos(20), Hohhot(41), Ulanqab(9), Baotou(49), Alxa League(6), Chifeng(12), Wuhai(7).
Ji Lin(267)	Changchun(179), Ji Lin(43), Tonghua(30), Liaoyuan(12), Bai-mountain(3).
Jiangsu(2365)	Nantong(238), Suzhou(560), Nanjing(545), Xuzhou(82), Wuxi(292), Changzhou(283), Yancheng(58), Yangzhou(105), Zhenjiang(68), Suqian(37), Lianyungang(40), Huaian(17), Taizhou(40)
Jiangxi(270)	Jingdezhen(25), Nanchang(128), Shangrao(28), Ganzhou(28), Yichun(14), Xinyu(23), Yingtan(7), Pingxiang(11), Jiu-River (1), Jian(2), Fuzhou(8).
Liaoning(536)	Panjin(13), Dalian(202), Shenyang(163), Benxi(11), Anshan(63), Fuxin(16), Liaoyang(18), Chaoyang, Fushun(12), Yingkou (7), Jinzhou(5), Dandong(2), Huludao(24).
Ningxia Hui Autonomous Region (75)	Yinchuan(40), Shizuishan(23), Central defender(12).
Qinghai(72)	Haidong(7), Xining(65).
Shaanxi(383)	Xian(292), Baoji(50), Xianyang(16), Shangluo(4), Hanzhong(11), Tongchuan(4), Yanan(6).
Shandong(1449)	Weifang(128), Jinan(211), Liaocheng(41), Binzhou(71), Zibo(231), Jining(58), Yantai(194), Taian(36), Linyi(36), Weihai (95), Dongying(49), Zaozhuang(9), Heze(10), Dezhou(44), Rizhao(6), Qingdao(230).
Shanghai (2014)	Shanghai (2014).
Shanxi(326)	Taiyuan(153), Yuncheng(45), Linfen(21), Xinzhou(11), Changzhi(26), Datong(28), Jincheng(11), Jinzhong(11), Yangquan (20).
Sichuan(870)	Chengdu(504), Deyang(29), Luzhou(27), Panzhihua(17), Suining(49), Neijiang(12), Mianyang(91), Nanchong(12), Yibin (29), Zigong(41), Yaan(10), Dazhou(5), Leshan(21), Guangan(4), Meishan(16), Aba Tibetan and Qiang Autonomous Prefecture(3).

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(*continued*)

Province/Municipality	City
Tianjin(409)	Tianjin(409).
Tibet Autonomous Region(50)	Lhasa(44), Nyingchi(6).
Xinjiang Uygur Autonomous Region(232)	Urumqi(186), Karamay(23), Changji Hui Autonomous Prefecture(6), The Ili Kazak Autonomous Prefecture(9), Bayingoleng Mongolian Autonomous Prefecture(8).
Yunnan(259)	Kunming(196), Lijiang(4), Qujing(32), Lincang(10), Yuxi(5), Diqing Tibetan Autonomous Prefecture(3), Zhaotong(5), Wenshan Zhuang and Miao Autonomous Prefecture(2).
Zhejiang(2612)	Hangzhou(915), Ningbo(477), Shaoxing(359), Jinhua(90), Lishui(27), Taizhou(299), Huzhou(179), Wenzhou(97), Jiaxing (162), Quzhou(31), Zhoushan(10).

Appendix B

Year and industry distribution of the sample.

This table summarizes the sample distribution across year and industry. Industry classification refers to the *Guidelines for Industry Classification of Listed Companies* issued by the China Securities Regulatory Commission in2012.[18]

Year															
Industry	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total
Agriculture, forestry, livestock farming, and fishery	4	4	9	10	19	18	18	21	23	25	29	32	31	35	278
Mining	11	10	18	29	40	46	51	58	66	70	64	65	69	74	671
Manufacturing[19]	324	412	564	749	1044	1138	1175	1177	1243	1576	1690	1761	1941	2079	16873
Electric power, gas and water production and supply	5	7	15	16	27	29	32	34	42	52	58	66	79	93	555
Construction	11	10	15	20	38	45	41	48	51	60	68	68	70	75	620
Wholesale and retail trade	3	3	4	12	34	37	37	46	49	57	65	72	83	94	596
Transport, storage and postal services	31	46	67	103	26	21	19	21	29	31	49	56	66	70	635
Accommodation and catering industry	4	8	17	23	1	2	2	2	2	4	5	5	5	5	85
Information transmission, software and information technology services	0	6	6	21	86	99	91	103	156	215	278	289	326	357	2033
Real estate	5	5	12	11	7	9	14	16	17	20	33	32	36	39	256
Leasing and business services	1	3	3	4	7	7	8	10	17	21	30	34	37	45	227
Scientific research and technical services	13	10	16	20	8	11	10	17	19	24	45	51	51	66	361
Water conservation, environment, and public facilities management	0	0	0	0	7	7	8	13	15	18	29	34	40	58	229
Education	0	0	0	0	1	1	1	1	1	1	7	4	5	9	31
Health and social work	0	0	0	0	2	2	3	4	5	5	7	8	9	9	54
Culture, sports, and entertainment	0	0	0	0	3	7	12	17	22	32	34	37	36	41	241
Comprehensive	0	0	0	0	8	10	11	12	14	13	11	13	10	7	109
Total	412	524	746	1018	1358	1489	1533	1600	1771	2224	2502	2627	2894	3156	23854

[19]According to the *Value Creation Report of China's Manufacturing Listed Companies in 2021*, as of December 31, 2021, the number of listed companies in China's manufacturing industry accounted for 66.7 % of the number of listed companies in China.

[18] See http://www.csrc.gov.cn/csrc/c100103/c1452025/content.shtml for more details.

Appendix C

Variable definitions.

This table presents the definitions of the main variables used in this paper.

	Variable	Definition
Dependent variables	$RD\_or_{i,t}$	The ratio of R&D investment to operating revenues multiplied by 100
	$RD\_asset_{i,t}$	The ratio of R&D investment to total assets multiplied by 100
Independent variables	$Cr_{i,t\text{-}1}$	Regional climate risk, the principal component analysis method was used to calculate the combined score for the secondary indicators after standardized and treated with absolute values, The greater the *Cr* is, the higher the regional climate risk is
Moderating Variable	$Cash_{i,t\text{-}1}$	The ratio of cash and marketable securities to total assets
	$Managshare_{i,t\text{-}1}$	The ratio of directors, officers and supervisors shares to total number of shares
Control variables	$Size_{i,t\text{-}1}$	Ln (total assets+1)
	$Age_{i,t\text{-}1}$	Ln (the number of years since the establishment time+1)
	$Debt_{,t\text{-}1}$	The ratio of total liabilities to total assets
	$Fixed_{i,t\text{-}1}$	The ratio of fixed assets to total assets
	$Tbq_{i,t\text{-}1}$	Market value/book value
	$AR_{i,t\text{-}1}$	(Total assets of the current period-Total assets of the previous period)/Total assets of the previous period
	$Fee_{i,t\text{-}1}$	Management expense/operating income
	$Ceo_{i,t\text{-}1}$	Dummy variable, assigned a value of 1 if the CEO and Chairman of the Board are the same individual; otherwise, assigned a value of 0
	$BS_{i,t\text{-}1}$	Number of the board of Supervisors
	$Hhi_{i,t\text{-}1}$	Sum of the squared ratio of the operating income of each company in the industry to the total operating income of the industry
	$Pgdp_{i,t\text{-}1}$	Total urban product divided by the number of people in the city (Unit 10000 yuan)
	$UP_{i,t\text{-}1}$	The ratio of the added value of the tertiary industry to that of the secondary industry in the city

Appendix D

The principal component structure of regional climate risk.

This table presents the principal component structure of regional climate risk of secondary indicators. The standardized secondary indicators are calculated by principal component analysis to measure climate risk.

1st-level Indicator	2nd-level Indicator	3rd-level indicator	Description	F1	F2	F3	F4	F5
Regional climate risk	Climate physical risk	Amt^{w}[20]	The annual average temperature of prefecture-level cities after standardization	−0.140	0.066	0.980	−0.127	−0.009
		Amp^{w}	Annual precipitation of prefecture-level cities after standardization	−0.564	−0.032	0.014	0.684	0.462
		$Amrh^{w}$	Annual average relative humidity of prefecture-level cities after standardization	0.545	0.130	0.159	0.715	−0.388
		Asd^{w}	Annual sunshine hours in prefecture-level cities after standardization	0.605	−0.150	0.099	−0.035	0.775
	Climate transition risk	IE^{w}	Total carbon dioxide emissions per unit GDP of prefecture-level cities	0.012	0.977	−0.072	−0.069	0.186

[20] Var^{w} is the secondary indicator after standardization, and the calculation and processing process are shown in Manuscript 3.2.2.

Appendix E

The difference test results of the impact of climate risk on firms' R&D investment between manufacturing and non-manufacturing industries.

	Panel A $RD\_or_{i,t}$			
	Fixed effects model		Tobit model	
	Manufacturing	Non-manufacturing	Manufacturing	Non-manufacturing
	(1)	(2)	(3)	(4)
***Cr*$_{i,t-1}$**	**−0.206***** **(-4.358)**	**−0.271***** **(-2.626)**	**−0.223***** **(-3.091)**	**−0.491**** **(-2.185)**
Size$_{i,t-1}$	0.073 (1.453)	0.107 (1.222)	0.040 (1.262)	−0.553*** (-6.220)
Age$_{i,t-1}$	−1.162*** (-7.384)	−0.855*** (-2.726)	−1.577*** (-15.838)	−1.401*** (-5.144)
Debt$_{,t-1}$	−2.206*** (-7.168)	−5.603*** (-9.345)	−2.957*** (-16.192)	−10.296*** (-18.613)
Fixed$_{i,t-1}$	−2.805*** (-7.582)	−2.648*** (-3.797)	−3.701*** (-14.815)	−6.487*** (-8.318)
Tbq$_{i,t-1}$	0.080* (1.829)	0.137 (1.475)	0.067*** (2.683)	0.179** (2.379)
AR$_{i,t-1}$	0.717*** (8.777)	0.536*** (2.919)	0.908*** (11.007)	0.535*** (2.626)
Fee$_{i,t-1}$	28.059*** (17.621)	23.294*** (11.273)	32.716*** (55.143)	33.491*** (26.223)
Ceo$_{i,t-1}$	0.367*** (3.834)	1.032*** (4.268)	0.460*** (6.833)	1.192*** (5.811)
BS$_{i,t-1}$	0.004 (0.079)	0.091 (1.193)	−0.041 (-1.314)	0.178* (1.875)
Hhi$_{i,t-1}$	−1.975*** (-5.728)	−3.364*** (-6.024)	−2.951*** (-11.618)	−3.156*** (-6.388)
Pgdp$_{i,t-1}$	0.112*** (7.169)	0.094*** (2.874)	0.115*** (11.091)	0.122*** (3.258)
UP$_{i,t-1}$	0.412*** (4.162)	0.207 (1.161)	0.439*** (5.713)	0.414* (1.830)
*Cons_*	3.493*** (3.187)	3.466 (1.626)	2.862*** (3.709)	18.326*** (8.103)
Year FE	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
Adj_R^2*/Pseudo R*^{2}	0.378	0.569	0.100	0.202
N	16873	6981	16873	6981
	Panel B $RD\_asset_{i,t}$			
	Fixed effects model		Tobit model	
	Manufacturing	Non-manufacturing	Manufacturing	Non-manufacturing
	(1)	(2)	(3)	(4)
***Cr*$_{i,t-1}$**	**−0.084***** **(-3.389)**	**−0.116***** **(-2.729)**	**−0.093**** **(-2.429)**	**−0.207**** **(-2.064)**
Size$_{i,t-1}$	−0.102*** (-3.290)	−0.125*** (-3.188)	−0.127*** (-7.626)	−0.455*** (-11.717)
Age$_{i,t-1}$	−0.497*** (-5.422)	−0.391*** (-2.773)	−0.693*** (-13.115)	−0.603*** (-4.946)
Debt$_{,t-1}$	−0.061 (-0.397)	−1.171*** (-4.788)	−0.342*** (-3.538)	−2.614*** (-10.647)
Fixed$_{i,t-1}$	−1.089*** (-5.153)	−1.120*** (-4.005)	−1.471*** (-11.143)	−2.689*** (-7.736)
Tbq$_{i,t-1}$	0.067*** (2.815)	0.098** (2.222)	0.063*** (4.683)	0.133*** (4.016)
AR$_{i,t-1}$	0.156*** (3.978)	0.082 (1.142)	0.249*** (5.669)	0.100 (1.091)
Fee$_{i,t-1}$	3.156*** (5.294)	4.360*** (4.868)	4.964*** (15.706)	7.805*** (13.583)
Ceo$_{i,t-1}$	0.049 (0.909)	0.110 (0.985)	0.089** (2.485)	0.126 (1.392)
BS$_{i,t-1}$	−0.079*** (-2.760)	0.013 (0.341)	−0.105*** (-6.416)	0.035 (0.838)

(*continued on next page*)

(*continued*)

	Panel A $RD\_or_{i,t}$			
	Fixed effects model		Tobit model	
	Manufacturing	Non-manufacturing	Manufacturing	Non-manufacturing
	(1)	(2)	(3)	(4)
$Hhi_{i,t-1}$	−1.168***	−1.665***	−1.663***	−1.618***
	(-5.960)	(-6.857)	(-12.347)	(-7.292)
$Pgdp_{i,t-1}$	0.055***	0.040***	0.056***	0.055***
	(5.721)	(2.942)	(10.234)	(3.248)
$UP_{i,t-1}$	0.152***	0.103	0.166***	0.191*
	(2.779)	(1.153)	(4.055)	(1.882)
*Cons_*	5.473***	5.520***	4.255***	12.249***
	(8.174)	(5.911)	(10.405)	(12.089)
Year FE	✓	✓	✓	✓
Ind FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
$Adj\_R^2/Pseudo\ R^2$	0.222	0.516	0.076	0.223
N	16873	6981	16873	6981

Note: *, **, *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively.

Data availability

Data will be made available on request.

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